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Research paper

Epidemiology-informed Spatiotemporal Graph Neural Network for heterogeneity-driven interpretable epidemic forecasting

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ABSTRACT

Accurate epidemic forecasting is crucial for effective disease control and prevention. Traditional mechanistic models often struggle to estimate epidemiological parameters that vary spatiotemporally, while deep learning-based approaches typically disregard intrinsic transmission dynamics and lack interpretability. To overcome these limitations, we propose a novel Epidemiology-informed Spatiotemporal Graph Neural Network (EISTGNN): a hybrid framework integrating the proposed Spatio-Contact Susceptible–Infectious–Recovered (SCSIR) model with a spatiotemporal graph neural network to model epidemic transmission dynamics across regions. The inherently smooth and continuous nature of interregional epidemic transmission indicates that consecutive graph structures share underlying latent patterns. To leverage this property, we employ an adaptive graph to model stable contact patterns, a temporal module to capture their fluctuating interactions, and fuse both into a spatiotemporal contact graph. To address the limitations of recurrent structures, we introduce a temporal decomposition module to extract long-term trends and short-term variations, which is then integrated with a spatiotemporal graph convolutional network to simultaneously identify epidemiological parameters and forecast outbreaks. We validate EISTGNN on real-world datasets at the provincial level in China and at the state level in Germany. Experimental results demonstrate that our method effectively models the spatiotemporal dynamics of infectious diseases, offering a valuable tool for epidemic modeling and forecasting. Furthermore, by analyzing the learned parameters through the effective reproduction number (R_t), we derive valuable insights into transmission mechanisms and enhance both interpretability and practical utility of the underlying model.

1. Introduction

The swift and widespread transmission of COVID-19, as characterized by its high contagion and significant morbidity, has profoundly affected global economies, international trade, healthcare systems, and, most importantly, countless lives (Baker et al., 2020). This pandemic has posed significant challenges to virtually all nations and regions, resulting in the implementation of various intervention tools to curb viral spread, such as lockdowns, the use of protective masks, and vaccination campaigns (Hsiang et al., 2020). Precise predictions of the pandemic's trajectory are essential for policymakers to devise effective containment strategies, optimize resource distribution, and strengthen healthcare infrastructure to protect populations. Nevertheless, the progression of an epidemic is inherently intricate, and traditional mathematical models often fail to adequately represent the complex and nonlinear behaviors observed in real-world scenarios. Fortunately, the significant progress in deep learning across disciplines such as image recognition, natural language processing, and big data analytics (Goodfellow et al., 2016; LeCun et al., 2015), has paved the way for its application in epidemic modeling as well (Yang et al., 2021a; Zhang et al., 2022; Wang et al., 2023), resulting in enhanced predictive accuracy and more reliable forecasts.

A variety of compartmental models, grounded in systems of differential equations, have been developed to simulate epidemic spread across populations; among these, the classical SIR model is foundational and offers inherent interpretability (Kermack and McKendrick,

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1927). These frameworks have been extended to incorporate realistic transmission characteristics and interventions – such as latent-period transmission, vaccination, and asymptomatic infections – thereby improving fidelity to real-world dynamics (Wallinga and Teunis, 2006; Nowak and May, 1999; Fisman et al., 2020). Given the pronounced temporal dependence of case data, forecasting is commonly cast as a time-series task: traditional methods like Autoregressive Integrated Moving Average (ARIMA) (Box and Jenkins, 1976a) and Support Vector Regression (SVR) (Smola and Schölkopf, 1998) provide statistical baselines, while recent deep learning models – including Recurrent Neural Networks (RNNs) (Rumelhart et al., 1986), Long Short-Term Memory networks (LSTM) (Hochreiter and Schmidhuber, 1997), Gated Recurrent Units (GRUs) (Cho et al., 2014), Temporal Convolutional Networks (TCNs) (Bai et al., 2018), Physics-Informed Neural Networks (PINNs) (Han et al., 2024; Raissi et al., 2019), and attention-based architectures (Transformer Vaswani et al., 2017, Temporal Fusion Transformer (TFT) Lim et al., 2020) – capture complex nonlinear patterns. However, these approaches predominantly emphasize temporal dynamics and often neglect spatial heterogeneity and contact patterns, which critically shape transmission and limit predictive robustness in realistic settings. This gap motivates integrated spatiotemporal approaches that combine the interpretability of mechanistic models with the representational power of data-driven learning.

In the development of such faithful spatiotemporal modeling frameworks, the spatial heterogeneity of populations emerges as a particularly critical and challenging dimension. The actual spread of infectious diseases is profoundly influenced by spatial variations in population density, contact behavior, and intervention policies across regions—factors often oversimplified or ignored. Traditional compartmental models, including the SIR framework and its variants, typically rely on a homogeneous mixing assumption (Smith and Doe, 2020; Johnson and Lee, 2019). This assumption fails to capture the complex, time-varying interactions between distinct sub-populations, leading to inaccurate predictions and potentially misguided control strategies (Garcia and Martinez, 2021). Consequently, integrating spatial heterogeneity is essential for faithful epidemic forecasting. This necessitates models that can simultaneously capture temporal dynamics and spatially-dependent interaction networks. Recent efforts have made significant strides in this direction: metapopulation models explicitly account for interregional mobility (Miller and Chen, 2022; Cao et al., 2023), network-based approaches refine contact pattern estimation (Davis and Kim, 2023), and machine learning techniques enhance spatial data integration (Lee and Park, 2023). While these approaches underscore the importance of spatial structure, they often rely on predefined or simplistic representations of inter-regional connections, which may not fully capture the dynamic and complex nature of real-world human interaction patterns.

The reliance on such pre-defined graphs constitutes a major limitation even for powerful Graph Neural Networks (GNNs), which are otherwise well-suited for modeling non-Euclidean spatiotemporal data (Wu et al., 2020b; Sun et al., 2023). While GNNs combine temporal processing with graph-based relational learning (Li et al., 2018), their effectiveness is critically dependent on the underlying graph structure. Prevailing approaches often resort to static graphs based on geographic proximity or heuristic risk encodings (Yu et al., 2018; Yang et al., 2021c), which inherently fail to adapt to temporal shifts caused by society-level influencing factors, such as holidays or policy interventions (Kraemer et al., 2020). In response, some studies have incorporated richer prior knowledge – such as mobility flows or gravity models – to construct weighted graphs (Li et al., 2018; Wu et al., 2020a). Yet, these remain approximations that are often biased and still cannot fully capture the complex, time-evolving nature of real-world interactions (Huang et al., 2020). A significant step forward comes from methods that learn adaptive adjacency matrices directly from data, such as MTGNN (Wu et al., 2020c), which offer dynamic flexibility. However, this focus on pure dynamics comes at a cost: the oversight of stable, long-term interaction patterns that are essential for mechanistic interpretability and reflect enduring societal structures. Consequently, the field faces a methodological trade-off: Current methods either prioritize stable structure at the expense of responsiveness to temporal variation, or emphasize dynamics while neglecting interpretable long-term regularities. This tension motivates approaches that integrate both aspects.

These limitations in graph learning are compounded by several deeper, interconnected challenges that stem from the long-standing divide between mechanistic and data-driven modeling paradigms. First, purely mechanistic approaches face a fundamental trade-off: model complexity quickly outpaces data availability, making it difficult to reliably estimate the numerous parameters required for detailed spatial models (Anderson and May, 1992; Heesterbeek et al., 2015). Second, while spatiotemporal GNNs (Wu et al., 2020b; Yu et al., 2018) and advanced hybrids like DSTAGNN (Li et al., 2022) attempt to combine graph components, they typically operate as black-box learners that lack any embedded epidemiological mechanism. This omission severs the link to domain theory and limits their ability to capture true interregional transmission dynamics (Kraemer et al., 2020; Wu et al., 2020a). Third, and as a consequence, the pursuit of predictive accuracy through deep learning often comes at the cost of mechanistic interpretability (Lipton, 2018; Rudin, 2019). Even methods offering post-hoc explanations, such as ColaGNN (Deng et al., 2019), fail to integrate explicit epidemiological principles or to disentangle stable structural dependencies from transient contextual fluctuations. In essence, the field is caught in a compromise: Models are either interpretable but limited, or powerful but opaque. This unresolved tension underscores the critical need for a new paradigm that moves beyond loose hybridization towards a truly cross-beneficial amalgamation of mechanistic modeling with state-of-art learning approaches. Such a framework must not only be computationally efficient but must also preserve – and indeed, derive – interpretability from its foundational principles (Karniadakis et al., 2021; Rackauckas et al., 2020).

Driven by these requirements, here we introduce the Epidemiology-informed Spatiotemporal Graph Neural Network (EISTGNN), a new framework explicitly designed to overcome the described limitations through a tight, bidirectional integration of mechanistic modeling and deep learning. Our approach is built on three key innovations that work in concert: (1.) a new spatiotemporal variant of the SIR model that explicitly quantifies inter-regional interactions via a contact rate matrix; (2.) a new spatiotemporal graph learner that co-learns both stable and dynamic graph structures to overcome the limitations of pre-defined adjacency matrices; (3.) a bidirectional closed loop between neural representation learning and mechanistic simulation, ensuring that the model's parameters and predictions are grounded in epidemiological principles. Integrating these components, our framework not only achieves state-of-the-art forecasting performance, but also provides two levels of mechanistic interpretability – from population-level contact parameters to macroscopic reproduction numbers – offering actionable, mechanism-aligned insights into transmission dynamics.

More in detail, to ground our framework in epidemiological theory, we (1.) develop an enhanced variant of the Spatio-Contact SIR (SCSIR) model: By introducing a region-pair-specific "contact rate" parameter, it explicitly quantifies inter-population interactions, transforming it from a standalone tool into a differentiable prior that is seamlessly embedded within the neural network. Crucially, the parameters of the SCSIR model (β, γ, c) are not merely intermediate variables; they act as bridge variables that create a tangible connection between the learned neural representations and the underlying epidemiological dynamics. (2.) To overcome the graph-learning dichotomy, we introduce a spatiotemporal graph learner that moves beyond the static-or-dynamic choice by co-learning an adaptive graph that captures persistent long-term contact patterns, and a dynamic temporal graph that models short-term, fluctuating interactions. This fusion provides a nuanced and evolving representation of spatial dependence that is both realistic and

efficient. Most importantly, we (3.) forge a bidirectional closed loop between data-driven learning and mechanistic simulation. This design is optimized through a closed-loop loss function that enforces consistency between the neural network's predictions and the trajectories simulated by the SCSIR model using the learned parameters. This process anchors the abstract features to epidemiological semantics, ensuring that the model remains faithful to domain knowledge. The neural network does not merely predict cases; it learns and outputs the epidemic parameters of the SCSIR model. These parameters are then fed into the SCSIR mechanics to generate a mechanistic forecast. The model is trained with a prediction loss and a mechanistic simulation loss, encouraging learned representations that are both accurate and epidemiologically consistent. Finally, a temporal decomposition module is incorporated to distill long-term trends from short-term variations, mitigating error accumulation and further enhancing the model's ability to capture complex temporal dynamics (Mao et al., 2024). Together, these components form a unified, end-to-end learning framework where mechanistic reasoning and representation learning mutually constrain and inform each other, enabling EISTGNN to deliver accurate, robust, and inherently interpretable forecasts.

We rigorously evaluate the performance of EISTGNN on two large-scale, real-world datasets: one encompassing COVID-19 spread across 31 provinces in China and another covering all 16 federal states in Germany. Comprehensive experiments demonstrate that our framework not only achieves superior forecasting accuracy over baselines across both short- and long-term horizons but also provides unique insights into the spatiotemporal dynamics of epidemic transmission. Furthermore, to provide mechanistic interpretability, we derive a time-varying effective reproduction number (R_t) from the learned parameters (β, γ, c) via a next-generation matrix, providing a coherent, decision-relevant summary of transmission potential across space and time. To assess predictive uncertainty, we performed multiple independent training runs. We observed high across-run reproducibility, indicating that the reported error bars are mainly attributable to training stochasticity. Overall, this analysis underscores the practical utility of our model for policy planning and offers a mechanistic interpretation of prediction results, thereby closing the loop between data-driven learning and epidemiological understanding.

This paper is organized as follows. Section 2 reviews the classical SIR model and introduces the proposed Spatio-Contact SIR model. Section 3 details the architecture of the proposed method. Section 4 reports simulated experiments and provides an in-depth analysis of the results. Finally, Section 5 concludes with a summary of contributions and a discussion of future directions.

2. The spatio-contact SIR model

Mathematical modeling of epidemiological dynamics is a well-established area of research in applied mathematics. The conventional compartmental models were developed to model the dynamics of epidemics within a population. A plain, yet powerful well-known compartmental model is the SIR model (Peeri et al., 2020), from which several models, such as SIRD and SEIR, originated. SIR assumes that the total population N remains constant. N is divided into three separate groups or compartments: susceptible (S), infectious (I) and removed (R), at each time t. Individuals are transferred between compartments as shown in Fig. 1 with certain rates, called transition rates, $\beta S(t)I(t)/N$ and $\gamma I(t)$: The simplest epidemiological compartmental models treat all individuals in the same compartment as sharing identical features. Therefore, each compartment is homogeneous. The SIR model is described by the following set of coupled ordinary differential equations:

$$\begin{aligned} \frac{dS(t)}{dt} &= -\frac{\beta I(t)}{N} S(t) , \\ \frac{dI(t)}{dt} &= \frac{\beta I(t)}{N} S(t) - \gamma I(t) , \\ \frac{dR(t)}{dt} &= \gamma I(t) . \end{aligned} \tag{1}$$

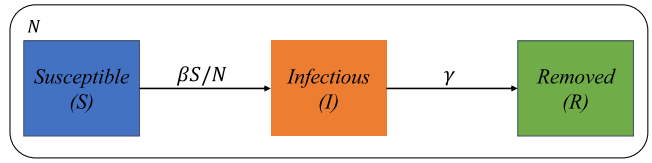

Fig. 1. Transition between compartments in the SIR model.

Here parameter β is the effective transmission rate. It denotes the number of effective contacts made by one infectious and one susceptible individual, which lead to one infection per unit of time. The removal rate γ indicates the fraction of infectious individuals who recover or die per unit of time. γ can be calculated using 1/D, with D being the average time duration that an infected individual can carry and transmit the virus. Eq. (1) is subject to the following initial conditions, $S(t_0) > 0, I(t_0) \geq 0,$ and $R(t_0) \geq 0$ at the initial time t_0. By construction, $S(t)+I(t)+R(t) = N$ holds at any time t. In general, the time scale of the epidemic dynamics is assumed to be short as compared to the length of the life of individuals in the population: the effects of births and natural deaths on the population are therefore not accounted for.

In the classical approach, the following steps are taken for parameter estimation and prediction in the SIR model:

I **Data Collection**: Gather data on infectious disease cases as observations.

II **Parameter Estimation**: Fit the model to the observed data by tuning β and γ, starting from the reasonable initial values. Minimizing the discrepancy between the model's predictions and the observed data provides the optimal (best fitting) parameter values.

III **Forecasting**: Use these optimized parameter values to predict future epidemic spread dynamics.

The original SIR model is too simple, but it offers a reasonable foundational framework for basic simulations of the dynamics of epidemic spread. However, the SIR model, with its oversimplified structure, has significant limitations. In particular, it struggles to account for population diversity and regional variation. In general, differences among individuals – such as age, gender, and other factors – have a significant impact on their behavior. This leads to an important variability in the resulting infection dynamics, known as the population diversity. Diversity is further subjected to regional differences: The simplest SIR model addresses primarily the spread of an epidemic within a single, homogeneous population or region, but neglects the interactions between different regions or sub-populations. In reality, epidemics often exhibit heterogeneous dynamics across various regions. This also involves complex interactions among regions, as illustrated in Fig. 2. To address these limitations, the Spatio-Contact SIR model, as detailed in Eq. (2), incorporates regional heterogeneity and contact patterns to better capture transmission dynamics across multiple regions.

Fig. 3 illustrates the key difference: the Spatio-Contact SIR model incorporates interregional interactions, whereas the classical SIR model assumes homogeneous mixing within a single region. While the SIR model tracks transitions between susceptible (S), infectious (I), and recovered (R) individuals within a single region, it does not account for the movement of individuals between different regions. In contrast, the Spatio-Contact SIR model categorizes the population of each region i into three compartments: $S_i(t)$ for susceptible individuals, $I_i(t)$ for infectious individuals, and $R_i(t)$ for recovered or deceased individuals in region i at time t. The total population size of region i, denoted by $N_i(t)$, is given by $N_i(t) = S_i(t) + I_i(t) + R_i(t)$. As in the SIR model, β represents the infection rate, while γ signifies the combined recovery and mortality rate. In addition, the Spatio-Contact SIR model

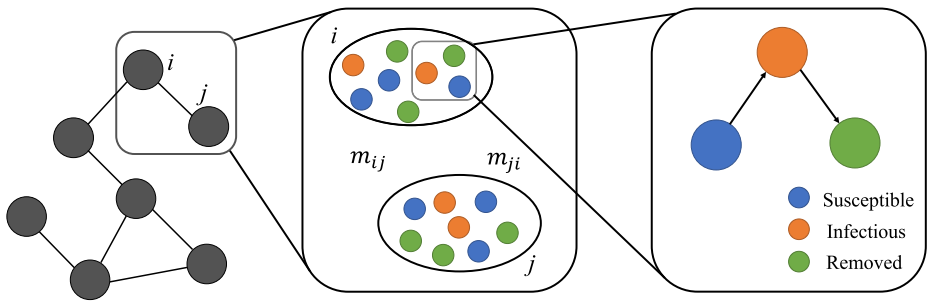

Fig. 2. Illustration of individual homogeneity and heterogeneity. Each black circle represents a distinct region, as shown in the center-left of the figure. The central part of the figure illustrates individual interregional movements, i and j. The right side of the figure depicts individuals transitioning between the S, I, and R types within a single region.

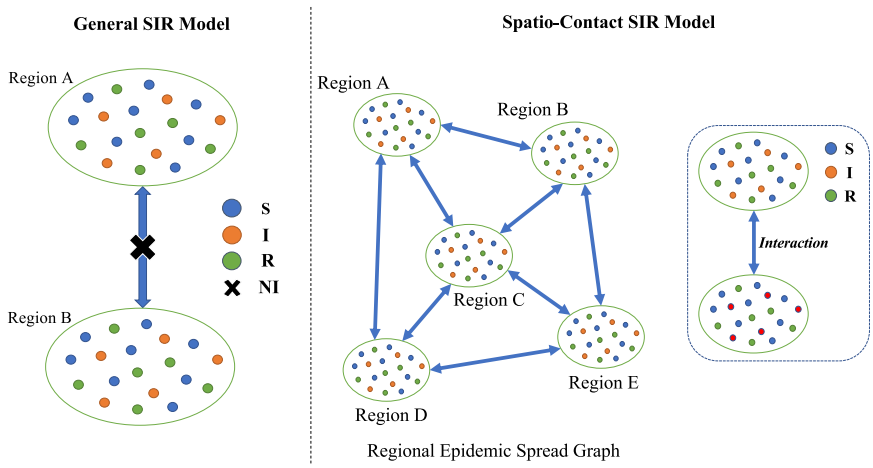

Fig. 3. Difference between the general SIR model and the Spatio-Contact SIR model, NI denotes no interaction between regions.

incorporates a contact rate c_{ij} to account for the intensity of epidemic transmission between region i and region j. The original Spatio-Contact SIR model is thus described by the following formulae:

$$\begin{aligned} \frac{dS_i(t)}{dt} &= -\beta \frac{S_i(t)}{N_i(t)} \sum_{j=1}^{Q} c_{ij} I_j(t), \\ \frac{dI_i(t)}{dt} &= \beta \frac{S_i(t)}{N_i(t)} \sum_{j=1}^{Q} c_{ij} I_j(t) - \gamma I_i(t), \\ \frac{dR_i(t)}{dt} &= \gamma I_i(t). \end{aligned} \tag{2}$$

Here, the parameter c_{ij} is used to construct an epidemic transmission graph (middle of Fig. 2), where $i, j \in \{1, \dots, Q\}$ index all regions. Realistic contact rates are generally positively correlated with interregional interaction intensities. Thus, by incorporating c_{ij}, the Spatio-Contact SIR model provides a more realistic simulation of disease spread both within and between regions, better reflecting actual transmission dynamics.

To facilitate integration with deep learning, the Spatio-Contact SIR model is discretized and parameterized, and is coupled with the EISTGNN framework via a bi-directional learning scheme, as detailed in Sections 3.3 and 3.6.

3. Methodology

In this section, the problem of spatiotemporal epidemic forecasting is defined. The overall framework of the proposed model is outlined and the specific details of each component are discussed.

3.1. Problem formulation

This study focuses on forecasting the number of infectious individuals for multiple regions and multiple time steps by using observed epidemic data.

Epidemic spread across different regions can be represented by a graph structure, denoted as $G(\mathcal{V}, \mathcal{E})$. Here $\mathcal{V}$ represents the set of regions and $\mathcal{E}$ denotes the set of weighted edges representing the interactions between these regions. This graph G can be transformed into an adjacency matrix $A \in \mathbb{R}^{Q \times Q}$, where each element A_{ij}, for $i, j \in \{1, \dots, Q\}$, indicates the edge weight between region i and region j, capturing the strength of transmission across these regions. The edge weight reflects the intensity of epidemic transmission across regions which varies over time, to account for the changes of the interactions.

These temporal dynamics are incorporated by a dynamic adjacency matrix, denoted as $\mathcal{A}_{1:T_{\text{obs}}} = [A_1, A_2, \dots, A_{T_{\text{obs}}}] \in \mathbb{R}^{T_{\text{obs}} \times Q \times Q}$, representing the evolution of regional connections over T_{obs} equidistant time steps. The spatiotemporal features of the epidemic data are then expressed as $\mathcal{X}_{1:T_{\text{obs}}} = [X_1, X_2, \dots, X_{T_{\text{obs}}}] \in \mathbb{R}^{T_{\text{obs}} \times Q \times F}$. Here each X_t for $t \in \{1, \dots, T_{\text{obs}}\}$ represents the F-dimensional historical epidemic data for Q regions at time step t. This dataset includes daily counts of the susceptible, infectious, removed (recovered and deceased) cases.

The goal of the spatiotemporal epidemic forecasting of this study is to leverage the historical epidemic data $\mathcal{X}$, along with the dynamic adjacency matrix $\mathcal{A}$, to learn a mapping function $f(\cdot)$. This function aims to predict the number of the infectious individuals for the Q regions over the T_{pre} future time steps, covering the period from $T_{\text{obs}} + 1$ to $T_{\text{obs}} + T_{\text{pre}}$, denoted as $Y_{T_{\text{pre}}} \in \mathbb{R}^{T_{\text{pre}} \times Q \times F}$. Thus, the epidemic

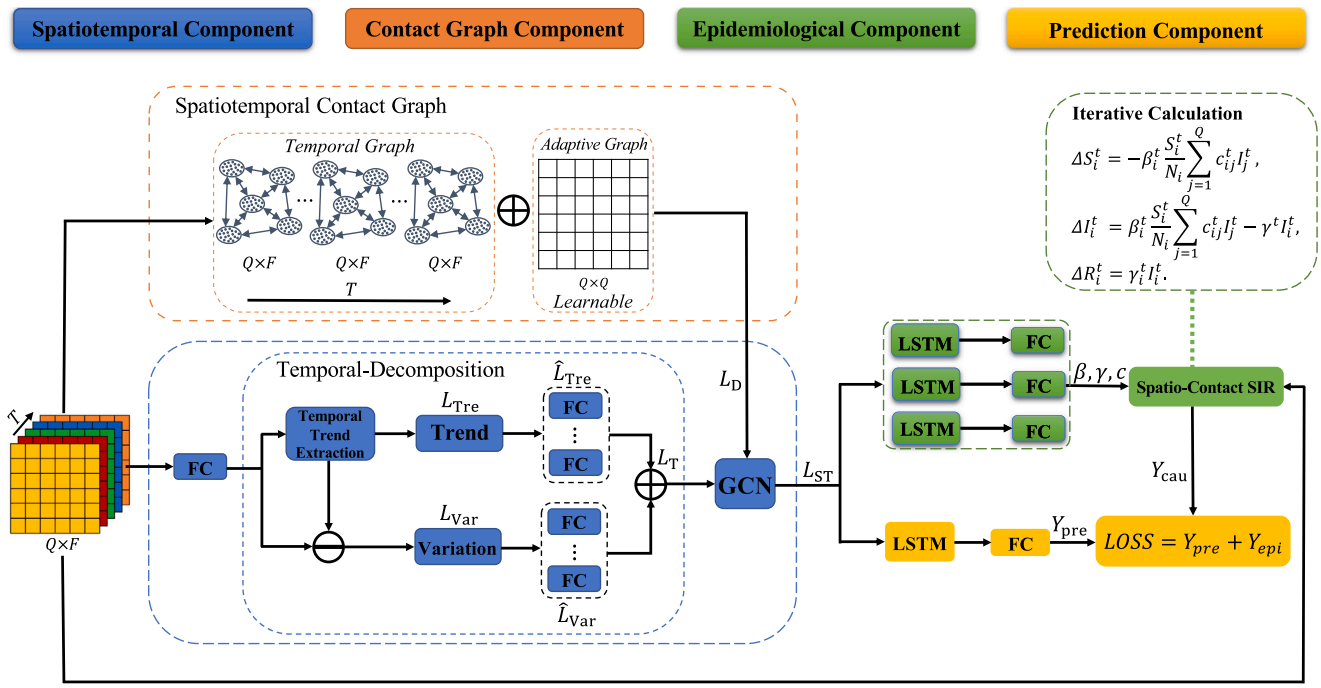

Fig. 4. The framework of the EISTGNN model.

forecasting can be formulated as follows:

$$\left\{\mathcal{X}_{1:T_{\text{obs}}}, \mathcal{A}_{1:T_{\text{obs}}}\right\} \xrightarrow{f(\cdot)} \mathbf{Y}_{T_{\text{obs}}+1:T_{\text{obs}}+T_{\text{pre}}} \tag{3}$$

3.2. Model overview

The overall architecture of the proposed EISTGNN is presented in Fig. 4: Four key components are integrated: the Spatiotemporal Component, the Contact Graph Component, the Epidemiological Component, and the Prediction Component. The Spatiotemporal Component employs a time decomposition model and a Graph Convolutional Network (GCN) to capture the spatiotemporal dynamics of epidemic data and predicts parameter sequences for the Spatio-Contact SIR model. The Contact Graph Component learns a spatiotemporal contact graph from epidemic data, which captures transmission dependencies across regions over time, and serves as a dynamic adjacency matrix input to the Spatiotemporal Component. The Epidemiological Component incorporates domain-specific knowledge to refine the model's representation of epidemic dynamics through learned epidemiological parameters. Notably, EISTGNN forms a closed-loop architecture in which mechanistic consistency guides the learning of spatiotemporal representations through residual feedback from epidemiological simulations. And finally, the Prediction Component combines neural network forecasts with epidemiological insights to jointly constrain predictions, thus enabling more reasonable multi-regional, multi-time-step forecasts of infectious individuals. The various parameters depicted in Fig. 4 are described in more detail in Table 1.

3.3. Discretization and parameterization of the spatio-contact SIR model

Although the SIR model is foundational in epidemiology, its homogeneous mixing assumption limits its capacity to capture regional transmission dynamics. In contrast, the Spatio-Contact SIR model incorporates population heterogeneity and advanced graph learning techniques to more accurately reflect the complex spatiotemporal spread of epidemics.

Table 1
Explanation of the parameters in Fig. 4.

Symbol	Description
T, Q, F	Temporal, regional, and feature dimensions of epidemic data
L_{Tre}	Trend dependence
L_{Var}	Variation dependence
$\ominus$	Element-wise subtraction by dimensional elements
$\hat{L}_{\text{Tre}}$	Trend dependence after feature-wise transformation
$\hat{L}_{\text{Var}}$	Variation dependence after feature-wise transformation
$\oplus$	Element-wise addition by dimensional elements
L_{T}	Temporal dependence
L_{D}	Dynamic adjacency dependence
L_{ST}	Spatiotemporal dependence
β	Predicted infection rate
γ	Predicted recovery rate
c	Predicted contact rates
Y_{pre}	Neural network prediction
Y_{epi}	Epidemiological inference

Here, interactions are modeled both within and between regional populations by using the Spatio-Contact SIR model. Accordingly, the contact rate c_{ij} in Eq. (2) is defined to represent the intensity of interregional interactions. Thus, greater interregional interaction intensity corresponds to higher contact rates. The spatiotemporal heterogeneity emerges in the dynamics of epidemic spread due to the influence of factors such as policies, weather, and their dynamic changes over time. To capture these variations, the model allows the infection rate β, the recovery rate γ, and the contact rate c_{ij} to vary across both space and time. To stabilize β and prevent it from becoming excessively small when multiplied by S_i^t, we simplify the calculation of S by merging β_i^t and S_i^t into a single term on the right-hand side of the equation for S. This adjustment assumes a relatively stable S value within each time step, which ensures numerical stability. To facilitate integration into a neural network framework, we discretize the Spatio-Contact SIR model in Eq. (2) using a first-order forward Euler method. For each compartment, the time derivative (e.g., $\frac{dS_i(t)}{dt}$) is approximated as $\frac{S_i^{t+1}-S_i^t}{\Delta t}$, and we set $\Delta t = 1$ for simplicity. This yields the discrete-time

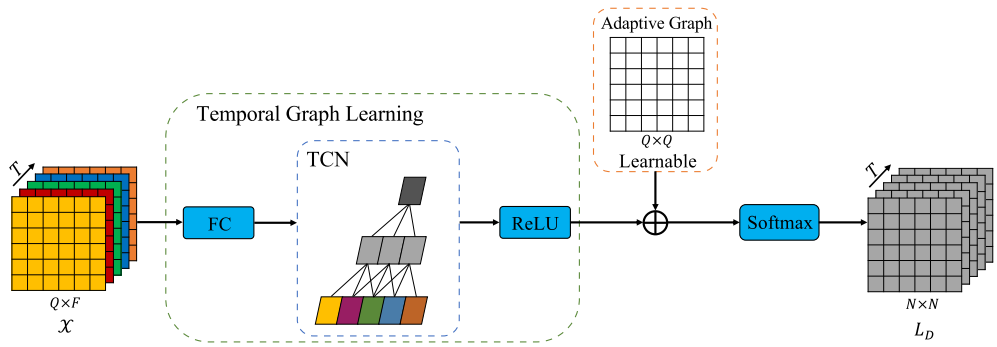

Fig. 5. Illustration of the spatiotemporal contact graph construction process. Input data is first processed by a fully connected (FC) layer, followed by a Temporal Convolutional Network (TCN) for temporal feature extraction. A ReLU activation is applied, and an adaptive graph refines the representation before the final Softmax output.

update equations:

$$\begin{aligned}\frac{S_i^{t+1}-S_i^t}{\Delta t} &= -\frac{\beta_i^t}{N_i^t}\sum_{j=1}^{Q} c_{ij}^t I_j^t,\\ \frac{I_i^{t+1}-I_i^t}{\Delta t} &= \frac{\beta_i^t}{N_i^t}\sum_{j=1}^{Q} c_{ij}^t I_j^t - \gamma_i^t I_i^t,\\ \frac{R_i^{t+1}-R_i^t}{\Delta t} &= \gamma_i^t I_i^t.\end{aligned} \tag{4}$$

Given the estimated parameters β_i^t, γ_i^t, and c_{ij}^t, the number of susceptible, infectious, and recovered individuals in each region can be iteratively computed using Eq. (4). Specifically, the compartmental states at time $t+1$ are obtained from those at time t as follows:

$$\left[S_i^t, I_i^t, R_i^t\right] \xrightarrow[\beta_i^t,\gamma_i^t,c_{ij}^t]{\text{Eq. (4)}} \left[S_i^{t+1}, I_i^{t+1}, R_i^{t+1}\right] \tag{5}$$

The specific parameter inference process is described in Section 3.6.

3.4. Contact graph component

Interregional interactions through individual contact fundamentally drive epidemic spread and shape transmission dynamics, exhibiting both long-term stable contact patterns and short-term fluctuations. To capture this duality, we employ two complementary graph representations as a spatiotemporal contact graph: an adaptive graph, which encodes slowly varying, stable connectivity pattern (e.g., geographic proximity and habitual mobility), and a temporal graph, which models rapid edge-weight changes usually induced by exogenous factors such as weather, policy interventions, and social events. Existing methods tend to focus on one view-adaptive graph approaches lack responsiveness to sudden shocks, while temporal graph techniques often ignore the stable interactions—leading to incomplete and inconsistent representations. Therefore, a unified framework that explicitly integrates adaptive and temporal graph learning is essential to fully capture the complex, static–dynamic nature of epidemic processes.

Furthermore, many spatiotemporal graph structures lack essential constraints, which results in significant structural discrepancies across successive time steps. These inconsistencies impair the ability of the graphs to reflect the smooth temporal evolution inherent in real-world epidemic processes and exacerbate the challenges of parameter optimization. In addition, these approaches frequently fail to encapsulate the latent interaction patterns that govern disease transmission, thereby limiting their effectiveness in accurately modeling and forecasting the spread of epidemics. Several studies (Mao et al., 2024; Yang et al., 2021b; Ye et al., 2022) suggest that the structure of dynamic graphs is typically continuous and smooth. Leveraging the similarity between adjacent time steps and incorporating temporal variation can help to generate dynamic graphs which can better reflect a realistic dynamical evolution.

Inspired by these observations, we construct an adaptive graph to capture the relatively stable contact patterns, thus reflecting the smooth nature of most daily activities. Simultaneously, we learn temporal graphs from epidemic data in order to model dynamic variations of interregional interaction patterns over time. We integrate these two components, by introducing constraints into the spatiotemporal contact graph learning process. This enables the model to better capture both stable and dynamic aspects of interregional interactions. This hybrid approach enhances the learning of contact graphs, providing an accurate representation of real-world interregional interaction during the spatiotemporal transmission of epidemics. The model's structure is shown in Fig. 5.

Adaptive graph structure have been widely used in the field of graph learning and can provide accurate adaptive graph information. Based on this perspective, we utilize the trainable embedding matrix $A_{\text{adaptive}} \in \mathbb{R}^{Q\times Q}$, where Q denotes the number of regions. The dynamic changes of a graph over time, can best be learned by mapping the epidemic data $\mathcal{X} \in \mathbb{R}^{T\times Q\times F}$ to a high-dimensional time-series embedding $A \in \mathbb{R}^{T\times Q\times F_{\text{T}}}$, using a fully connected layer. Here F and F_{T} represent the dimensions before and after the mapping:

$$A = FC(\mathcal{X}), \tag{6}$$

this embedding is then fed into a temporal model, to learn the temporal features. It is also passed through the spatiotemporal module for further processing.

Temporal Convolutional Networks (TCNs) are widely recognized for their ability to capture long-range temporal dependence in sequential data. By leveraging causal and dilated convolutions, TCNs ensure that the output at each time step depends only on the current and the previous steps, preserving temporal causality. In addition, TCNs support parallel computation and enable efficient training, thus making them particularly suitable for time-series data.

Given the embedding $A \in \mathbb{R}^{T\times Q\times F_{\text{T}}}$, TCN processes A along the temporal dimension T. The operation in a single TCN layer can be expressed as:

$$A_{\text{temporal}}(t,q,f) = \sum_{i=0}^{K-1}\sum_{f'=1}^{F_{\text{T}}} W_{i,f',f}\cdot A(t-d\cdot i,q,f') + b_f \tag{7}$$

Here $W \in \mathbb{R}^{K\times F_{\text{T}}\times F_{\text{TCN}}}$ is the trainable convolution kernel, and $b \in \mathbb{R}^{F_{\text{TCN}}}$ is the bias term. Here, f indexes the output feature channels after the convolution operation, K is the kernel size, and d represents the dilation rate. F_{T} and F_{TCN} denote the input and output feature dimensions, respectively.

By stacking multiple TCN layers, the final output, $A_{\text{temporal}} \in \mathbb{R}^{T\times Q\times F_{\text{TCN}}}$, is obtained. This representation captures effectively both short-term and long-term temporal dependence, and facilitates downstream spatiotemporal modeling.

To construct a spatiotemporal contact graph which combines both stable and dynamic information, the stable adaptive graph is merged

with the learned temporal graph. To better integrate the adaptive graph A_{adaptive} and the dynamic graph A_{temporal}, a fully connected (FC) layer to A_{temporal} is applied. This does not only transform A_{temporal} into a form that matches the shape and dimensions of A_{adaptive}, but it also enhances the integration of temporal features into the stable structure. Next, an element-wise addition of the adaptive graph A_{adaptive} and the transformed temporal graph $FC(A_{\text{temporal}})$ is performed. Numerical stability and interpretability is ensured by applying the $Softmax$ function to the fused graph for normalization, thus mapping its values to the range from 0 to 1. The resulting dynamic graph $L_{\text{D}} \in \mathbb{R}^{T\times Q\times Q}$ therefore represents the interaction weights between regions over time. The steps for the calculation are as follows:

$$L_{\text{D}} = Softmax\left(A_{\text{adaptive}} \oplus FC\left(A_{\text{temporal}}\right)\right), \tag{8}$$

where $\oplus$ denotes element-wise (entrywise) addition. Fusing the adaptive graph with the temporal graph captures both stable and transient inter-regional interaction patterns. This design yields a structurally coherent spatiotemporal contact graph that provides a useful inductive bias for downstream modeling, thereby reducing reliance on additional structural regularizers. The resulting graph L_{D} thus serves as a learned, time-varying adjacency matrix encoding evolving inter-regional interactions for subsequent graph-convolutional operations (see Section 3.5). Importantly, L_{D} is a representational construct for feature propagation within the GCN and is distinct from the mechanistic contact rate $c_{ij}(t)$ used in the Spatio-Contact SIR model. The model subsequently infers the interpretable epidemiological parameters from the spatiotemporal representations shaped by L_{D} (Section 3.6).

3.5. Spatiotemporal component

The Spatiotemporal Component is designed to jointly capture the temporal and spatial transmission dynamics of epidemics. For the temporal modeling module, a time-decomposition approach is adopted, which is widely used in time series forecasting research. That approach focuses on decomposing epidemic data into trends and variations along the temporal dimension. In this way, error accumulation is effectively mitigated and robust learning of temporal dependence is ensured. Spatial features are modeled using a GCN widely used in graph learning research, and integrated with the learned temporal graph in Section 3.4. Combining the extracted temporal and spatial features, this framework provides a comprehensive understanding of both, the temporal evolution and spatial interactions in epidemic data. Hence, a reliable basis for capturing the spatiotemporal spread of infectious diseases is provided.

To begin, the mapping of epidemic data $\mathcal{X} \in \mathbb{R}^{T\times Q\times F}$ to a high-dimensional time-series embedding $L \in \mathbb{R}^{T\times Q\times F_{\text{T}}}$ is used, employing a fully connected layer. Here F and F_{T} represent the dimensions before and after the mapping,

$$L = FC(\mathcal{X}) \tag{9}$$

The Temporal-Decomposition Module is designed to decompose epidemic data along the temporal dimension into two distinct features, Trend and Variation. Since epidemic time series are typically dominated by low-frequency dynamics, a simple moving average is adopted to extract the trend component, offering a computationally efficient and interpretable solution (Rivieccio et al., 2021). This process begins with the Temporal Trend Extraction Block, which identifies the overarching temporal trends present in the input data. The trend component $L_{\text{Tre}} \in \mathbb{R}^{T\times Q\times F_{\text{T}}}$ captures the smooth and long-term evolution of the data. This is computed using a convolution operation with a special kernel, which is designed with fixed and equal weights that sum to 1. This ensures that all values within the temporal window contribute equally to the output. Thus, a smooth and consistent representation of the underlying trend is provided. The variation component $L_{\text{Var}} \in \mathbb{R}^{T\times Q\times F_{\text{T}}}$ isolates short-term fluctuations from dynamic patterns. It is calculated as the element-wise difference between L_{Tre} and the input data. Both, L_{Tre} and L_{Var}, are then processed through fully connected layers to learn refined representations. Thus outputs $\hat{L}_{\text{Tre}} \in \mathbb{R}^{T\times Q\times F_{\text{T}}}$ and $\hat{L}_{\text{Var}} \in \mathbb{R}^{T\times Q\times F_{\text{T}}}$ retain their respective temporal features. Finally, the reconstructed output L_{T} integrates these learned representations. This ensures that the decomposed components effectively describe the temporal dynamics of the epidemic data. This decomposition process not only reduces error accumulation, but also enhances the robustness of downstream modeling tasks. The corresponding calculation formulas are

$$\begin{aligned}(L_{\text{Tre}}, L_{\text{Var}}) &= TemporalDecomposition(L),\\ L_{\text{Tre}} &= MovingTrendExtraction(Padding(L)),\\ L_{\text{Var}} &= L \ominus L_{\text{Tre}}\end{aligned} \tag{10}$$

Here $TemporalDecomposition()$ refers to the process of decomposing the time series into its trend and variation components. $MovingTrend\ Extraction()$ represents the moving average operation, applied with padding, to handle boundary effects. This ensures that all time steps are included in the computation. $\ominus$ represents element-wise subtraction.

The trend L_{Tre} and variation L_{Var} components, extracted through temporal decomposition, are processed by fully connected layers, applied separately to each feature along the feature dimension. These operations produce updated temporal embedding representations, $\hat{L}_{\text{Tre}}$ and $\hat{L}_{\text{Var}}$, corresponding to the trend and variation components, respectively. Finally, the updated trend and variation components are aggregated through an addition operation to produce the final embedding representation, L_{T}, which effectively captures temporal dependence. The computation process can be described as follows:

$$\begin{aligned}\hat{L}_{\text{Tre}} &= FC_{feature}(L_{\text{Tre}}),\\ \hat{L}_{\text{Var}} &= FC_{feature}(L_{\text{Var}}),\\ L_{\text{T}} &= \hat{L}_{\text{Tre}} \oplus \hat{L}_{\text{Var}}.\end{aligned} \tag{11}$$

In addition to temporal dependence, the spread of epidemics is significantly influenced by spatial interactions among regions. To incorporate this spatial impact, the temporal embedding $L_{\text{T}} \in \mathbb{R}^{T\times Q\times F_{\text{T}}}$ and the dynamic adjacency graph $L_{\text{D}} \in \mathbb{R}^{T\times Q\times \text{Q}}$ are utilized, which are learned according to Section 3.4, as inputs to the GCN model. The GCN aggregates spatiotemporal information by modeling the interactions between regions through the adjacency relationships encoded in L_{D} and by the temporal features in L_{T}. Specifically, the dynamic graph L_{D}, which represents normalized spatial connections, is applied to L_{T} in order to propagate information across connected regions. This transmission is followed by a linear transformation parameterized by W. This method generates an embedding representation, $L_{\text{ST}} \in \mathbb{R}^{T\times Q\times F_{\text{T}}}$, effectively capturing both temporal and spatial dependence. The computation process is:

$$\begin{aligned}L_{\text{ST}} &= GCN(L_{\text{D}}, L_{\text{T}})\\ &= ReLU(L_{\text{D}} L_{\text{T}} W + b)\end{aligned} \tag{12}$$

Here $W \in \mathbb{R}^{T\times Q\times F_{\text{T}}}$ represents the weight parameters, b denotes the bias term, and $ReLU$ is the activation function. This framework allows the GCN to learn how spatial interactions and temporal dependence jointly influence the spread of epidemics. This makes the resulting representation L_{ST} well-suited for downstream predictive tasks.

3.6. Epidemiological component

Among epidemic researches, there is a growing consensus that relying solely on spatiotemporal models is insufficient for higher accurately forecasting infectious disease spreading dynamics. These models typically capture only the patterns and distributions inherent in the data, often lacking the explicit incorporation of fundamental physical rules which govern epidemic dynamics. Given the extensive array of domain knowledge models available in epidemiological research, integrating these models into neural networks as prior knowledge could enhance the networks' ability to better grasp the underlying physical

principles of epidemic dynamics. Such integration promises to render deep learning-based epidemic predictions better interpretable and more robust.

While many recent hybrid frameworks use GNNs solely to regress epidemiological parameters such as β and γ, our EISTGNN introduces a deeper integration by enforcing a mechanistic constraint on the spatiotemporal features. Specifically, the spatiotemporal embedding L_{ST}, extracted via temporal decomposition and GCN, is used to infer β_t^i, γ_t^i, and $c_{ij}(t)$ through a neural network predictor. These inferred parameters are subsequently fed into the discretized Spatio-Contact SIR model to generate epidemic state updates. Crucially, the discrepancy between these physically predicted trajectories and the ground-truth epidemic data is explicitly minimized through the joint loss function Eq. (18). This process forms a closed loop that allows domain knowledge to inform, constrain, and regularize the feature extraction process within the graph-learning pipeline. As a result, the interaction between deep learning and physical modeling becomes bi-directional, ensuring that EISTGNN does not merely learn parameters but integrates mechanistic reasoning throughout its training.

To integrate epidemiological context, this component incorporates the mechanistic differential equations from the Spatio-Contact SIR model (see Section 3.3) into the neural network framework: In the Spatio-Contact SIR model, N_i denotes the population of individuals in region i, each of whom can be in one of the following states: S, I, R. Compartmental models operate under a homogeneous mixing assumption, i.e., every infected individual can directly infect any other individual. The dynamics of epidemic spread in region i at time t are described by the following equations:

$$\begin{aligned} \Delta S_i^t &= -\Delta t \cdot \beta_i^t \frac{S_i^t}{N_i^t} \sum_{j=1}^{Q} c_{ij}^t I_j^t, \\ \Delta I_i^t &= \Delta t \cdot \left(\beta_i^t \frac{S_i^t}{N_i^t} \sum_{j=1}^{Q} c_{ij}^t I_j^t - \gamma_i^t I_i^t \right), \\ \Delta R_i^t &= \Delta t \cdot \gamma_i^t I_i^t \end{aligned} \tag{13}$$

Here S_i^t, I_i^t, and R_i^t denote the number of individuals in each state within region i at time t, with the constraint $S_i^t + I_i^t + R_i^t = N_i^t$. The parameters β_i^t, γ_i^t, and c_{ij}^t represent the infection rate, the removal rate, and the contact rate between regions i and j, respectively. In this framework, these parameters are considered mechanistically interpretable quantities across Q regions at time t, and they are inferred using a neural network. The term Δ refers to the newly added number of individuals in each state.

By estimating these mechanistically-grounded parameters using a neural network, the Spatio-Contact SIR model into the neural network framework can be integrated. Specifically, the spatiotemporal dependence of epidemic spread, as captured by the component $L_{\mathrm{ST}} \in \mathbb{R}^{T_{\mathrm{obs}} \times N \times F}$, is put into a neural network. This neural network is designed to learn both the infection rates, removal rates, and contact rates $\beta, \gamma \in \mathbb{R}^{T_{\mathrm{pre}} \times N \times 1}$, $c \in \mathbb{R}^{T_{\mathrm{pre}} \times N \times N}$, which govern epidemic dynamics. These rates are then normalized using the Sigmoid function to ensure they remain within reasonable bounds.

$$\begin{aligned} \beta &= Sigmoid(FC(LSTM(Reshape(L_{\mathrm{ST}})))), \\ \gamma &= Sigmoid(FC(LSTM(Reshape(L_{\mathrm{ST}})))), \\ c &= Sigmoid(FC(LSTM(Reshape(L_{\mathrm{ST}})))) \end{aligned} \tag{14}$$

Here $Sigmoid()$ refers to the activation function, $LSTM()$ represents the Long Short-Term Memory network (LSTM), which is used to capture temporal dependence in the data and to make predictions for future time steps. $FC()$ denotes the fully connected layer in the neural network, and $Reshape()$ changes the shape of the data.

Finally, the outbreak's evolution across all regions can be predicted by combining the last observed epidemic data $(S_{T_{\mathrm{obs}}}, I_{T_{\mathrm{obs}}}, R_{T_{\mathrm{obs}}})$ with the estimated parameters β, γ, and c from the Spatio-Contact SIR model and by applying Eq. (13):

$$(\Delta S_{1:T_{\mathrm{pre}}}^{\mathrm{epi}}, \Delta I_{1:T_{\mathrm{pre}}}^{\mathrm{epi}}, \Delta R_{1:T_{\mathrm{pre}}1}^{\mathrm{epi}}) = SCSIR((S_{T_{\mathrm{obs}}:-1}, I_{T_{\mathrm{obs}}:-1}, R_{T_{\mathrm{obs}}:-1}), \beta, \gamma, c) \tag{15}$$

Here $SCSIR$ refers to the Spatio-Contact SIR model, and $(\Delta S_{1:T_{\mathrm{pre}}}^{\mathrm{epi}}, \Delta I_{1:T_{\mathrm{pre}}}^{\mathrm{epi}}, \Delta R_{1:T_{\mathrm{pre}}}^{\mathrm{epi}}) \in \mathbb{R}^{T_{\mathrm{pre}} \times N \times 1}$ denote the changes in the numbers of susceptible, infectious, and removed individuals across all regions over T_{pre} time steps. This closed-loop mechanism not only ensures consistency with epidemiological dynamics but also constrains the learned spatiotemporal contact graph via mechanistic feedback. Furthermore, the SCSIR model serves as an additional inductive bias, guiding the evolution of the spatiotemporal contact graph towards meaningful and temporally coherent structures.

3.7. Prediction component

As illustrated in Fig. 4, the entire framework is constrained by two outputs: neural network output and mechanistic output. The prediction from the neural network output is used as our final prediction, as it embeds the hidden information from all components.

The output of the neural network is generated by feeding the feature L_{ST}, as learned by the Spatiotemporal Component, which captures the dynamics of the epidemic spread, into a fully connected layer with an activation function. This approach enables prediction of the epidemic's temporal evolution, specifically the number of infectious individuals Y_{pre}, across all regions for the subsequent T_{pre} time steps:

$$Y_{1:T_{\mathrm{pre}}}^{\mathrm{pre}} = ReLU(FC(LSTM(Reshape(L_{\mathrm{ST}})))) \tag{16}$$

Here $FC()$ and $LSTM()$ serve the same functions as introduced in Eq. (14), and $ReLU()$ denotes the activation function.

The output of the mechanistic is obtained by iteratively applying Eq. (13). This process utilizes the estimated epidemiological parameters to run the Spatio-Contact SIR model. Thus, the mechanistic prediction for the number of infectious individuals is generated across all regions for the next T_{pre} time steps:

$$\begin{aligned} (S_1^{\mathrm{epi}}, I_1^{\mathrm{epi}}, R_1^{\mathrm{epi}}) &= (S_{T_{\mathrm{obs}}:-1}, I_{T_{\mathrm{obs}}:-1}, R_{T_{\mathrm{obs}}:-1}) + (\Delta S_1^{\mathrm{epi}}, \Delta I_1^{\mathrm{epi}}, \Delta R_1^{\mathrm{epi}}), \\ (S_2^{\mathrm{epi}}, I_2^{\mathrm{epi}}, R_2^{\mathrm{epi}}) &= (S_1^{\mathrm{epi}}, I_1^{\mathrm{epi}}, R_1^{\mathrm{epi}}) + (\Delta S_2^{\mathrm{epi}}, \Delta I_2^{\mathrm{epi}}, \Delta R_2^{\mathrm{epi}}), \\ &\vdots \\ (S_{T_{\mathrm{pre}}}^{\mathrm{epi}}, I_{T_{\mathrm{pre}}}^{\mathrm{epi}}, R_{T_{\mathrm{pre}}}^{\mathrm{epi}}) &= (S_{T_{\mathrm{pre}}-1}^{\mathrm{epi}}, I_{T_{\mathrm{pre}}-1}^{\mathrm{epi}}, R_{T_{\mathrm{pre}}-1}^{\mathrm{epi}}) + (\Delta S_{T_{\mathrm{pre}}}^{\mathrm{epi}}, \Delta I_{T_{\mathrm{pre}}}^{\mathrm{epi}}, \Delta R_{T_{\mathrm{pre}}}^{\mathrm{epi}}), \end{aligned} \tag{17}$$

Here $(S_{1:T_{\mathrm{pre}}}^{\mathrm{epi}}, I_{1:T_{\mathrm{pre}}}^{\mathrm{epi}}, R_{1:T_{\mathrm{pre}}}^{\mathrm{epi}}) \in \mathbb{R}^{T_{\mathrm{pre}} \times N \times 1}$ represents the mechanistic prediction for each state at the T_{pre} time steps, and $Y_{1:T_{\mathrm{pre}}}^{\mathrm{epi}} \in \mathbb{R}^{T_{\mathrm{pre}} \times N \times 1}$ $(= I_{1:T_{\mathrm{pre}}}^{\mathrm{epi}})$ indicates the projected number of infectious individuals for the following T_{pre} time steps, in accord with the Spatio-Contact SIR model.

3.8. Optimization

The proposed approach is optimized by minimizing the discrepancy between the predicted quantities and the ground truth by using a loss function based on the Mean Absolute Error (MAE), formulated as follows:

$$L(\theta) = \frac{1}{Q \times T} \sum_{q=1}^{Q} \sum_{t=1}^{T} \left(|Y_{q,t}^{\mathrm{pre}} - Y_{q,t}^{\mathrm{obs}}| + |Y_{q,t}^{\mathrm{epi}} - Y_{q,t}^{\mathrm{obs}}| \right) \tag{18}$$

Here θ denotes the unknown trainable parameters, Q denotes the total number of regions, T represents the number of predicted future time steps, and Y^{pre} and Y^{epi} are the outputs of the forecasting component and the epidemic inference component, respectively. Y^{obs} is the ground truth. This loss function constrains the training process by minimizing the difference between the outputs of both, the forecasting component Y^{pre} and the epidemic inference component Y^{epi}, from the ground truth Y^{obs}. The average errors across all regions are calculated and prediction time steps. It provides a comprehensive measure of the prediction accuracy, for both components.

Table 2
Statistical information of each dataset.

Dataset	Timeframe	Size	Max	Min	Mean	Std Dev
China-Provinces	2021.10.30~2022.10.30	$365 \times 31 \times 3$	126009300	0	15 158 910	27561640
Germany-States	2021.10.30~2022.10.30	$365 \times 16 \times 3$	17 175 680	560	1 763 944	2848876

4. Experiments

4.1. Datasets

We utilized two distinct real-world datasets related to epidemics for our experiments: the China-Provinces dataset, which includes data from 31 selected Chinese provinces, and the Germany-States dataset, covering all 16 German federal states. Each dataset provides daily cumulative counts for each region r – confirmed $C_t^{(r)}$, recoveries $R_t^{\text{rec},(r)}$, and deaths $D_t^{(r)}$ – together with the regional population $N^{(r)}$. From these series we derive the compartments: $S_t^{(r)} = N^{(r)} - C_t^{(r)}$, $I_t^{(r)} = C_t^{(r)} - R_t^{\text{rec},(r)} - D_t^{(r)}$, and $R_t^{(r)} = R_t^{\text{rec},(r)} + D_t^{(r)}$, so that $S_t^{(r)} + I_t^{(r)} + R_t^{(r)} = N^{(r)}$.

Statistical information about the datasets is presented in Table 2, further details are provided below

- **China-Provinces**: These 31 province-level datasets were collected from DXY (Blanker, 2024), which provides data from the Chinese Center for Disease Control and Prevention (CCDC). These datasets contain province-level population data and COVID-19-related records in China, covering the period from October 30, 2021, to October 30, 2022 (365 days).
- **Germany-States**: These 16 state-level datasets were collected by the Robert Koch Institute (RKI) (Robert Koch-Institut, 2024) and include population data and COVID-19-related records for each of the 16 states of Germany over the period from October 30, 2021, to October 30, 2022 (365 days).

These two datasets span a full year for each country, covering all seasons and major holidays. This comprehensive timeframe allows us to capture seasonal variations and interregional interaction patterns, both of which significantly influence epidemic spread. Moreover, the stark contrast between China's vast population and Germany's medium-sized population – together with their differing control policies – introduces additional heterogeneity in transmission dynamics, enabling us to evaluate our model's generalization across diverse epidemic scenarios. In our experiments, we use active cases as the primary feature, with susceptible and removed cases serving as auxiliary features.

4.2. Comparison models

To evaluate performance in predicting epidemic spread across 31 provinces in China and 16 states in Germany, we compare EISTGNN with five categories of baselines: (i) traditional methods: SIR (Susceptible–Infectious–Removed), SEIR (Susceptible–Exposed–Infectious–Removed), SCSIR (Spatio-Contact SIR), ARIMA (Autoregressive Integrated Moving Average); (ii) traffic spatiotemporal models: Graph WaveNet (GraphWaveNet); (iii) epidemic spatiotemporal models: Multi-Scale Spatio-Temporal Graph Neural Network (MSGNN); (iv) transformer-based models: Spatio-Temporal Graph Transformer (STGformer); (v) adapted models: Spatio-Temporal Graph Convolutional Network (STGCN) and Spatio-Temporal Graph Neural ODE (STGODE). These baseline models will be evaluated against our proposed EISTGNN in Section 4.5 to assess their performance across different forecasting horizons and datasets.

(1) **SIR** (Kermack and McKendrick, 1927): The SIR model as one of the most fundamental compartmental models in epidemiology accounts for weekly periodicity, if the optimized values of the infection rate β and removal rate γ from the previous week is used to generate the predictions.

(2) **ARIMA** (Box and Jenkins, 1976b): The ARIMA is a classical statistical model that forecasts a univariate time series by combining autoregression, differencing, and moving-average components.

(3) **SEIR**: The SEIR model extends the classical SIR framework by adding an 'Exposed' compartment to represent the latent period between infection and infectiousness, enabling more realistic dynamics for diseases with incubation. Parameters are calibrated on historical data and then used for forward simulation and forecasting.

(4) **SCSIR**: The SCSIR model considers regional heterogeneity, and it models interactions between regions. For each region, the optimized β, γ, and contact matrix c from the previous week is used to produce the predictions.

(5) **GraphWaveNet** (Wu et al., 2019): GraphWaveNet constructs an adaptive graph via learnable node embeddings and combines diffusion Graph Convolutional Networks (GCNs) with gated Temporal Convolutional Networks (TCNs) to capture spatiotemporal dependencies.

(6) **MSGNN** (Qiu et al., 2024): The MSGNN learns graphs at multiple geographic scales (e.g.,micro/macro), applies both scale-specific and scale-shared graph convolutions, and aggregates cross-scale temporal dependencies to produce multi-step forecasts. Unlike fixed-graph baselines, it jointly learns graph structure and performs multi-scale message passing before the forecasting head.

(7) **STGformer** (Wang et al., 2024): The STGformer integrates Transformer architectures with graph learning for spatiotemporal forecasting. Through self-attention, it dynamically models temporal dependencies across time steps and spatial correlations between nodes, capturing global spatiotemporal context for improved prediction.

(8) **STGCN**: The STGCN (Yu et al., 2018) combines graph convolution with temporal convolution for spatiotemporal forecasting. For a fair and rigorous comparison, we substitute its fixed adjacency matrix with our dynamic graph learning module (Section 3.4), while keeping all other components unchanged.

(9) **STGODE**: The STGODE framework (Fang et al., 2021) couples graph neural networks with ODE-based continuous-time dynamics to form a spatiotemporal graph neural ODE. For a fair and rigorous comparison, we substitute its fixed adjacency matrix with our dynamic graph learning module (Section 3.4) and adapt the output head to estimate compartmental-model parameters that drive a mechanistic simulator for prediction, while retaining all other original components.

4.3. Setup of experiments

We split each dataset chronologically into training, validation, and test sets with ratios of 60%–20%–20%. All data are normalized to the range $(0, 1)$ via per-region population-based normalization. The input sequence length is set to 7 days, motivated by the characteristic serial interval of SARS-CoV-2 (Ali et al., 2020). We evaluate forecasting performance at output horizons of 7 and 14 days (short term) and 21 and 28 days (long term), thereby covering both short- and longer-range planning scenarios. We construct overlapping sliding windows with stride = 1, ensuring that no window crosses the predefined split boundaries. For a window starting at time t, the first 7 days serve as the input and the subsequent k days ($k \in \{7, 14, 21, 28\}$) are the prediction targets.

Hyperparameters were selected via a structured validation sweep under a fixed computational budget, evaluating options for learning rate ({1e−3, 1e−4, 1e−5}), hidden dimension ({16, 32, 64}), and depth ({2, 3} layers). Following a parsimony principle, we selected the smallest effective configuration performing within 1% of the best validation MAE. The final configuration – a hidden dimension of 16 for both the temporal-decomposition module and the 3-layer GCN, and a 1-layer TCN with kernel size 3 – is shared across both datasets. All tuning was performed exclusively on the validation set, keeping the test set strictly unseen until the final evaluation.

We employ a curriculum learning strategy that progressively increases the prediction horizon from one-day-ahead to the target k-ahead, and use early stopping (patience=50) with a maximum of 200 epochs per horizon. The model is optimized using Adam with a learning rate of 1×10^{-4} and a mini-batch size of 8. All experiments are implemented in PyTorch and conducted on two NVIDIA A100 GPUs. Code is available at https://github.com/ichbinshhan/Epi-Spatio-Temporal-Teilen.

4.4. Evaluation metrics

Several metrics are employed to evaluate the performance of the proposed model: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Relative Absolute Error (RAE), Pearson Correlation Coefficient (PCC), and Concordance Correlation Coefficient (CCC). These metrics assess both, the accuracy of the predictions (MAE, RMSE, RAE) and the degree of agreement between the predicted and observed values (PCC, CCC). Low values of MAE, RMSE, and RAE indicate higher prediction accuracy, while higher values of PCC and CCC suggest a stronger correlation with the ground truth. To evaluate the influence of randomness (e.g., from random initialization), we conduct five independent training runs for each model and report the mean values along with the empirical 95% uncertainty intervals. For temporal outputs (forecast trajectories and the effective reproduction number R_t), we plot the mean curve alongside its empirical 95% uncertainty band, aggregated from the ensemble of runs at each time point. These intervals and bands primarily reflect the variability due to training stochasticity and are presented consistently across all figures and tables in follow sections. The formal definitions of these evaluation metrics are provided here.

MAE quantifies the average magnitude of the prediction errors, disregarding their direction:

$$MAE = \frac{1}{Q \times T} \sum_{q=1}^{Q} \sum_{t=1}^{T} \left| Y_{q,t}^{\text{pre}} - Y_{q,t}^{\text{obs}} \right| \tag{19}$$

Here $Y_{q,t}^{\text{pre}}$ and $Y_{q,t}^{\text{obs}}$ represent the predicted and observed values at location q and time t, respectively.

RMSE is the square root of the average squared differences between predictions and observations, emphasizing larger errors:

$$RMSE = \sqrt{\frac{1}{Q \times T} \sum_{q=1}^{Q} \sum_{t=1}^{T} \left(Y_{q,t}^{\text{pre}} - Y_{q,t}^{\text{obs}} \right)^2}. \tag{20}$$

RAE measures the relative magnitude of the total absolute error compared to the total variation of the observed data:

$$RAE = \frac{\sum_{q=1}^{Q} \sum_{t=1}^{T} \left| Y_{q,t}^{\text{pre}} - Y_{q,t}^{\text{obs}} \right|}{\sum_{q=1}^{Q} \sum_{t=1}^{T} \left| Y_{q,t}^{\text{obs}} - \bar{Y}^{\text{obs}} \right|} \tag{21}$$

Here $\bar{Y}^{\text{obs}}$ is the mean of the observed values.

PCC assesses the linear correlation between predicted and observed values:

$$PCC = \frac{\sum_{q=1}^{Q} \sum_{t=1}^{T} \left(Y_{q,t}^{\text{pre}} - \bar{Y}^{\text{pre}} \right) \left(Y_{q,t}^{\text{obs}} - \bar{Y}^{\text{obs}} \right)}{\sqrt{\sum_{q=1}^{Q} \sum_{t=1}^{T} \left(Y_{q,t}^{\text{pre}} - \bar{Y}^{\text{pre}} \right)^2} \sqrt{\sum_{q=1}^{Q} \sum_{t=1}^{T} \left(Y_{q,t}^{\text{obs}} - \bar{Y}^{\text{obs}} \right)^2}} \tag{22}$$

Here $\bar{Y}^{\text{pre}}$ and $\bar{Y}^{\text{obs}}$ are the means of the predicted and observed values, respectively.

CCC extends PCC by considering both precision and accuracy, measuring the agreement between predicted and observed values:

$$CCC = \frac{2\,\rho\,\sigma_{\text{pre}}\,\sigma_{\text{obs}}}{\sigma_{\text{pre}}^2 + \sigma_{\text{obs}}^2 + \left(\mu_{\text{pre}} - \mu_{\text{obs}} \right)^2} \tag{23}$$

Herein ρ is the Pearson Correlation Coefficient as calculated above, while σ_{pre} and σ_{obs} are the standard deviations, and μ_{pre} and μ_{obs} are the means of the predicted and observed values, respectively. Unlike PCC, CCC accounts for both the correlation and the mean squared difference between the predictions and observations, providing a comprehensive measure of agreement. The CCC ranges from −1 to +1, where +1 indicates perfect concordance and −1 indicates total discordance.

Employing these metrics thoroughly evaluates the performance of all models in our subsequent comparisons from multiple perspectives, which ensures a robust assessment of their predictive capabilities.

4.5. Performance evaluation

This experiment compares the EISTGNN model with the four baseline models for short-term and long-term epidemic forecasting. The results on the China and Germany datasets are summarized in Tables 3 and 4. Bold values indicate the best performance for each metric; underlined values indicate the second-best. "Improvement" reports the relative gain of EISTGNN over the second-best model for that metric, computed as |best − second|/second (interpreted as error reduction for MAE/RMSE/RAE and percentage increase for PCC/CCC). A dash (–) is shown when EISTGNN is not the best, or when the best result is tied. The EISTGNN model generally achieves the best or the most competitive performance, across different forecasting tasks and datasets, as compared to all other models. In what follows, we present time-series visualizations under the best-performing setting indicated by Tables 3 and 4, focusing on our model and the strongest-performing baselines as a concise qualitative complement to the quantitative results.

On the China dataset, shown in Table 3, the EISTGNN model achieves the best overall performance across all forecasting horizons. In short-term forecasting (L = 7,14), EISTGNN demonstrates significant gains over the strongest baseline (STGODE), with an 11.41% reduction in MAE, 12.38% in RMSE, and a 1.03% increase in CCC at L = 7. At L = 14, it remains leading in MAE (and RAE), while RMSE is led by MSGNN and CCC by STGODE, with EISTGNN ranking closely second in these cases. This indicates that EISTGNN is highly competitive even against recent strong baselines. In long-term forecasting (L = 21,28), EISTGNN continues to deliver strong results: at L = 21 it improves MAE by 8.54%, RMSE by 12.47%, RAE by 16.67%, and CCC by 2.17%; at L = 28 it still reduces MAE by 1.33% and RMSE by 2.24% and improves CCC by 1.38% over the runner-up, while PCC is topped by STGformer and RAE by MSGNN with small margins. Overall, these results underscore the robustness and consistency of EISTGNN on China, with occasional metric crossovers driven by models specialized for specific error or correlation criteria.

On the Germany dataset, illustrated in Table 4, the EISTGNN delivers either the best or highly competitive performance across all horizons. In short-term forecasting (L = 7,14), it achieves improvements of 13.1%/9.8% in MAE and 9.6%/9.2% in RMSE, together with 2.5%/5.3% in CCC, compared to the second-best model. In long-term forecasting, its superiority is pronounced at L = 21 with 11.0% (MAE) and 17.3% (RMSE) gains and a 3.9% increase in CCC; at L = 28 it further improves RMSE by 8.4% and CCC by 3.5% and remains competitive in MAE (0.4%), while PCC is marginally higher for STGODE. Taken together, these findings support the superiority and stability of EISTGNN for long-term prediction on Germany, with minor exceptions on individual metrics that are consistent with metric-specific sensitivities.

Table 3
Performance comparison with baseline models on the China dataset.

	The China dataset									
	L = 7					L = 14				
Model	MAE	RMSE	RAE	PCC	CCC	MAE	RMSE	RAE	PCC	CCC
SIR	77.1 ± 1.0	250.2 ± 7.5	0.4 ± 0.0	89.6 ± 0.8%	75.8 ± 1.8%	102.8 ± 0.8	340.8 ± 5.5	0.6 ± 0.0	75.5 ± 1.5%	53.7 ± 1.8%
SEIR	103.3 ± 0.3	326.1 ± 2.4	0.6 ± 0.0	73.2 ± 0.6%	57.8 ± 0.8%	169.7 ± 0.7	385.5 ± 3.8	0.7 ± 0.0	60.0 ± 1.1%	40.8 ± 1.8%
ARIMA	149.2 ± 9.7	459.5 ± 17.6	0.8 ± 0.1	68.3 ± 1.3%	70.4 ± 1.2%	213.9 ± 10.8	471.8 ± 18.1	0.8 ± 0.2	75.5 ± 1.5%	46.3 ± 1.0%
SCSIR	79.5 ± 1.1	261.2 ± 7.5	0.4 ± 0.0	88.9 ± 1.1%	72.9 ± 1.8%	104.5 ± 0.5	349.0 ± 2.8	0.6 ± 0.0	74.2 ± 0.7%	50.7 ± 1.0%
GraphWaveNet	54.9 ± 4.4	95.3 ± 15.0	0.5 ± 0.0	89.1 ± 0.5%	84.1 ± 0.5%	77.1 ± 15.0	155.5 ± 56.9	0.5 ± 0.1	83.9 ± 4.6%	83.3 ± 4.5%
MSGNN	51.0 ± 4.0	99.7 ± 9.5	0.4 ± 0.1	89.1 ± 0.3%	79.0 ± 0.3%	86.8 ± 7.7	**137.7 ± 23.9**	0.4 ± 0.1	93.4 ± 1.6%	85.9 ± 1.5%
STGformer	47.9 ± 2.7	97.4 ± 4.8	0.3 ± 0.0	88.8 ± 0.2%	97.1 ± 0.2%	63.4 ± 1.9	144.5 ± 4.7	0.4 ± 0.2	78.1 ± 0.2%	87.8 ± 0.2%
STGCN	53.2 ± 6.3	186.6 ± 12.9	0.5 ± 0.0	88.2 ± 0.6%	77.9 ± 0.7%	92.0 ± 18.7	155.1 ± 51.8	0.5 ± 0.1	94.3 ± 3.9%	94.0 ± 3.9%
STGODE	44.7 ± 9.9	101.6 ± 28.0	0.3 ± 0.1	97.5 ± 1.5%	96.9 ± 1.7%	65.1 ± 5.0	143.6 ± 11.7	0.4 ± 0.0	95.4 ± 0.8%	**94.5 ± 1.1%**
EISTGNN	**39.6 ± 5.3**	**83.5 ± 12.1**	**0.2 ± 0.0**	**98.3 ± 0.5%**	**98.1 ± 0.5%**	**58.5 ± 10.3**	147.7 ± 27.2	**0.3 ± 0.1**	**95.4 ± 1.3%**	93.8 ± 3.6%
Improvement	11.41%	12.38%	33.33%	0.82%	1.03%	7.73%	–	25.0%	0.05%	–
	L = 21					L = 28				
Model	MAE	RMSE	RAE	PCC	CCC	MAE	RMSE	RAE	PCC	CCC
SIR	110.6 ± 1.1	330.1 ± 4.8	0.6 ± 0.0	71.0 ± 1.0%	49.3 ± 2.2%	114.5 ± 1.0	324.7 ± 4.9	0.7 ± 0.0	68.9 ± 1.4%	47.9 ± 2.0%
SEIR	213.0 ± 0.6	362.0 ± 2.9	0.7 ± 0.0	58.0 ± 1.2%	39.8 ± 1.4%	324.6 ± 0.5	353.3 ± 2.8	0.7 ± 0.0	56.8 ± 1.2%	38.9 ± 1.8%
ARIMA	247.2 ± 11.4	437.0 ± 18.7	0.8 ± 0.1	39.7 ± 1.2%	40.4 ± 1.1%	345.4 ± 12.9	422.5 ± 17.2	0.8 ± 0.2	50.6 ± 1.5%	30.5 ± 1.0%
SCSIR	111.6 ± 0.8	334.4 ± 6.7	0.6 ± 0.0	69.9 ± 2.1%	47.5 ± 2.6%	116.1 ± 0.5	333.2 ± 2.2	0.7 ± 0.0	67.1 ± 0.5%	44.0 ± 1.2%
GraphWaveNet	107.9 ± 15.0	251.4 ± 57.9	0.6 ± 0.1	78.4 ± 8.7%	80.4 ± 8.4%	143.5 ± 30.5	385.1 ± 136.8	0.8 ± 0.2	64.8 ± 14.3%	62.4 ± 15.6%
MSGNN	99.5 ± 14.7	243.5 ± 46.6	0.6 ± 0.1	81.2 ± 4.0%	79.4 ± 4.5%	138.5 ± 5.5	277.1 ± 14.4	**0.5 ± 0.2**	71.9 ± 1.4%	71.1 ± 1.3%
STGformer	114.0 ± 7.5	293.7 ± 19.7	0.7 ± 0.2	83.1 ± 1.7%	82.9 ± 1.7%	113.2 ± 3.6	336.5 ± 9.6	0.7 ± 0.0	**84.5 ± 1.0%**	72.5 ± 1.0%
STGCN	102.6 ± 31.3	271.2 ± 91.8	0.7 ± 0.2	76.0 ± 12.5%	73.8 ± 14.8%	127.5 ± 46.8	324.0 ± 157.4	0.7 ± 0.2	78.3 ± 11.3%	55.2 ± 13.9%
STGODE	100.1 ± 14.3	243.6 ± 54.5	0.6 ± 0.1	84.4 ± 7.3%	82.8 ± 7.3%	127.6 ± 43.2	322.4 ± 146.8	0.7 ± 0.3	75.8 ± 14.3%	72.3 ± 15.9%
EISTGNN	**91.0 ± 16.6**	**213.1 ± 31.8**	**0.5 ± 0.1**	**88.3 ± 4.7%**	**84.7 ± 7.7%**	**111.7 ± 44.1**	**270.9 ± 100.9**	0.6 ± 0.3	75.0 ± 23.8%	**73.5 ± 23.0%**
Improvement	8.54%	12.47%	16.67%	4.62%	2.17%	1.33%	2.24%	–	–	1.38%

Table 4
Performance comparison with baseline models on the Germany dataset.

	The Germany dataset									
	L = 7					L = 14				
Model	MAE	RMSE	RAE	PCC	CCC	MAE	RMSE	RAE	PCC	CCC
SIR	8527.7 ± 44.2	15264.1 ± 84.7	0.5 ± 0.0	90.5 ± 0.2%	75.3 ± 0.3%	10233.8 ± 76.6	18135.4 ± 168.5	0.6 ± 0.0	83.6 ± 0.4%	62.1 ± 0.9%
SEIR	10837.9 ± 67.5	18649.3 ± 105.4	0.6 ± 0.0	86.1 ± 0.1%	62.1 ± 0.5%	11658.6 ± 43.7	20012.6 ± 81.1	0.7 ± 0.0	81.2 ± 0.2%	53.3 ± 0.4%
ARIMA	14503.8 ± 24.0	23713.7 ± 43.0	0.8 ± 0.1	80.2 ± 0.1%	38.3 ± 0.2%	13854.2 ± 19.0	22701.5 ± 26.0	0.8 ± 0.1	80.4 ± 0.1%	39.7 ± 0.5%
SCSIR	8531.0 ± 65.0	15276.6 ± 152.8	0.5 ± 0.0	90.4 ± 0.3%	75.3 ± 0.6%	10286.4 ± 13.1	18170.2 ± 72.6	0.6 ± 0.0	83.6 ± 0.3%	61.9 ± 0.4%
GraphWaveNet	6493.7 ± 178.6	11250.1 ± 318.1	0.4 ± 0.1	92.9 ± 4.3%	87.0 ± 7.0%	9761.8 ± 279.7	17247.8 ± 483.0	0.5 ± 0.2	76.4 ± 20.6%	67.3 ± 21.2%
MSGNN	5810.7 ± 131.8	9443.7 ± 193.6	0.4 ± 0.0	95.8 ± 2.5%	90.6 ± 3.9%	8800.8 ± 132.7	13957.9 ± 203.6	0.6 ± 0.1	89.2 ± 5.0%	70.0 ± 6.2%
STGformer	5309.8 ± 393.5	8901.6 ± 200.3	0.4 ± 0.1	96.6 ± 0.4%	92.0 ± 1.1%	8557.9 ± 361.0	14344.3 ± 537.3	0.6 ± 0.0	90.3 ± 0.5%	76.7 ± 1.8%
STGCN	7447.2 ± 142.7	13069.1 ± 259.3	0.4 ± 0.1	90.9 ± 1.4%	82.2 ± 1.1%	8951.3 ± 168.5	15923.7 ± 311.8	0.5 ± 0.2	84.6 ± 1.6%	70.8 ± 1.3%
STGODE	5055.0 ± 393.2	8308.9 ± 323.0	0.3 ± 0.1	97.2 ± 1.9%	92.1 ± 4.3%	8208.1 ± 667.2	13846.8 ± 469.6	0.5 ± 0.1	92.2 ± 5.4%	75.5 ± 16.0%
EISTGNN	**4393.8 ± 416.8**	**7511.6 ± 184.2**	**0.2 ± 0.1**	**97.7 ± 1.3%**	**94.4 ± 2.5%**	**7400.7 ± 949.6**	**12577.1 ± 361.6**	**0.4 ± 0.1**	**92.7 ± 1.9%**	**80.8 ± 11.1%**
Improvement	13.1%	9.6%	33.3%	0.5%	2.5%	9.8%	9.2%	20.0%	0.5%	5.3%
	L = 21					L = 28				
Model	MAE	RMSE	RAE	PCC	CCC	MAE	RMSE	RAE	PCC	CCC
SIR	11605.3 ± 42.3	20122.5 ± 73.3	0.7 ± 0.0	80.1 ± 0.2%	52.6 ± 0.4%	11991.0 ± 41.6	20614.2 ± 62.7	0.7 ± 0.0	78.3 ± 0.2%	47.9 ± 0.4%
SEIR	12448.2 ± 22.1	21105.7 ± 42.1	0.7 ± 0.0	79.5 ± 0.1%	47.7 ± 0.3%	12543.3 ± 18.8	21152.5 ± 29.9	0.7 ± 0.0	78.8 ± 0.2%	45.3 ± 0.2%
ARIMA	13836.2 ± 176.9	22663.0 ± 324.5	0.8 ± 0.1	80.5 ± 1.4%	39.8 ± 1.0%	13511.9 ± 169.2	22060.3 ± 312.7	0.8 ± 0.2	81.0 ± 1.5%	40.7 ± 1.1%
SCSIR	11614.2 ± 66.8	20119.3 ± 131.0	0.7 ± 0.0	80.2 ± 0.3%	52.5 ± 0.7%	12008.1 ± 74.2	20593.2 ± 110.2	0.7 ± 0.0	78.4 ± 0.3%	48.1 ± 0.6%
GraphWaveNet	12413.1 ± 365.1	20697.6 ± 121.7	0.7 ± 0.2	73.6 ± 15.5%	69.3 ± 14.4%	17012.9 ± 237.8	28704.4 ± 109.5	0.8 ± 0.4	74.7 ± 35.1%	74.4 ± 31.7%
MSGNN	9614.8 ± 272.3	15848.1 ± 316.7	0.5 ± 0.2	85.7 ± 3.5%	79.7 ± 4.6%	9687.9 ± 184.8	16844.6 ± 334.3	0.6 ± 0.1	78.4 ± 5.4%	73.8 ± 7.5%
STGformer	8414.4 ± 138.1	14833.9 ± 301.7	0.6 ± 0.0	85.8 ± 0.6%	82.8 ± 1.0%	9754.5 ± 240.1	17495.3 ± 308.1	0.6 ± 0.0	74.5 ± 1.5%	64.4 ± 1.2%
STGCN	10300.5 ± 153.4	18028.2 ± 287.9	0.6 ± 0.2	80.6 ± 1.5%	61.7 ± 1.2%	10769.5 ± 171.6	18735.5 ± 303.2	0.6 ± 0.1	78.1 ± 1.4%	56.7 ± 1.3%
STGODE	9229.0 ± 473.7	14772.3 ± 633.3	0.5 ± 0.1	87.9 ± 2.3%	80.6 ± 8.1%	8460.7 ± 823.0	15159.5 ± 664.3	**0.5 ± 0.1**	**87.5 ± 3.2%**	71.3 ± 10.6%
EISTGNN	**7487.5 ± 425.9**	**12220.5 ± 931.9**	**0.4 ± 0.1**	**89.0 ± 3.4%**	**86.0 ± 4.1%**	**8423.9 ± 422.2**	**13880.9 ± 1406.2**	**0.5 ± 0.1**	86.2 ± 3.7%	**77.0 ± 15.7%**
Improvement	11.0%	17.3%	20.0%	1.3%	3.9%	0.4%	8.4%	–	–	3.5%

Moreover, informed by the results in Tables 3 and 4, we fix the visualization horizon at L = 7 days, which attains strong overall performance across models and datasets. To create a unified view of the model's 7-day-ahead forecasts across the entire test set, we plot, for each day τ, the prediction made 7 days earlier ($\hat{y}_{\tau|\tau-7}$). These predictions form a continuous timeline because the testing windows overlap. Connecting them yields a single curve that shows the model's projected trend throughout the test period. To demonstrate the efficacy of this forecast curve, we apply it to specific regions. We select four representative provinces from the China dataset and four typical states from the Germany dataset to clearly illustrate the comparison between forecasted and reported active cases in a 7-day ahead prediction (L = 7), with a focus on the highest-performing models. Fig. 6 displays the forecasting curves for Beijingshi (2022-08-19 to 2022-10-30), Chongqingshi, Shaanxisheng, and Guizhousheng in China. Fig. 7 shows the forecasting curves for Berlin (2022-08-19 to 2022-10-30), Hessen, Bayern, and Thüringen in Germany, respectively. Observe that the EISTGNN model's forecasting curves are generally smoother and closer to the ground truth, across both datasets. For instance, the EISTGNN model accurately fits the upward trend of active cases, nearly overlapping with the reported values during the period from 2022-09-15 to 2022-09-30 in Guizhousheng and Hessen. Furthermore, as time progresses, the EISTGNN model shows superior performance in capturing long-term trends, in particular in the later stages (e.g., 2022-10-10 to 2022-10-30). Take, for example, Shaanxisheng and Thüringen: here, the EISTGNN model successfully captures the upwards or stabilizing trends in active cases during that period, whereas other models exhibit rather large deviations.

These results demonstrate that the EISTGNN model does not only capture the dynamic trends efficiently and effectively in short-term

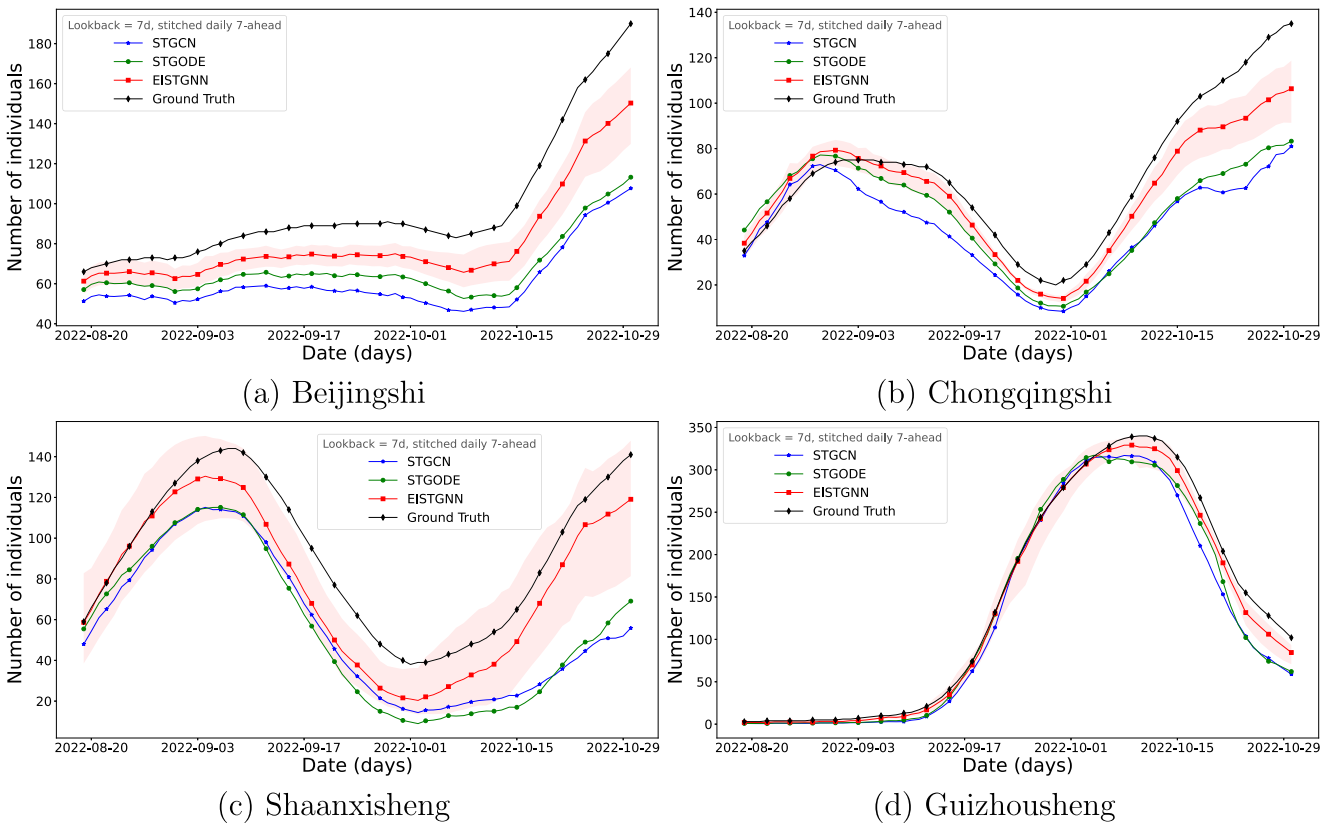

Fig. 6. Visualization of 7-day-ahead forecasts for active cases in representative provinces of China. For each calendar date τ, the value $\hat{y}_{\tau|\tau-7}$ represents the prediction made 7 days prior. The continuous curve is formed by connecting these daily predictions across the test period. The ground truth is shown in black, comparisons from other models in color, and the shaded band represents the 95% interval over an ensemble of 5 independent training runs, reflecting prediction uncertainty due to model stochasticity.

forecasting but also excels in long-term predictions, by showing higher robustness and better accuracy by adapting to the evolving trends. Other models, in contrast, tend to diverge significantly when the reported values exhibit abrupt changes, e.g. the declining phase around 2022-10-01 in Chongqingshi. This observation is consistent with the results in Tables 3 and 4, further confirming the EISTGNN model's superiority in forecasting tasks, particularly its ability to predict trends accurately as the forecasting horizon increases.

Next, the performance of different models is directly compared, traditional mechanistic models perform poorly. SIR cannot capture complex spatiotemporal dependencies or adaptively estimate time-varying parameters, causing accuracy to degrade over longer horizons. While SEIR adds an exposed compartment, noise and nonstationarity hinder parameter identifiability, often causing it to underperform even SIR. ARIMA, lacking any spatial coupling, struggles profoundly with nonstationary epidemic data, making it the weakest baseline. Although SCSIR incorporates time-varying interregional contact rates, it remains constrained by its simplified, preset assumptions, limiting its ability to learn dynamic, complex interactions.

Non-mechanistic deep learning models demonstrate the value of data-driven learning but exhibit clear limitations. GraphWaveNet and MSGNN surpass traditional models by learning spatiotemporal patterns, yet their general-purpose architectures (e.g., dilated TCNs, multiscale graphs) omit epidemiological priors (e.g., compartmental flows), leading to performance ceilings and instability—e.g., MSGNN leads in RMSE at L = 14 and RAE at L = 28 on China but not in correlation metrics, while GraphWaveNet shows high short-term RMSE but significant variance. Similarly, STGformer achieves high long-horizon PCC (L = 28) yet often at the cost of elevated MAE/RMSE, reflecting the trade-offs of its global attention bias without mechanistic grounding.

Domain-informed models deliver the strongest performance. For a rigorous comparison, we adapted STGCN and STGODE by integrating our dynamic graph learner (Section 3.4). The improved performance of STGCN confirms the value of a dynamic graph, but its lack of embedded epidemiological mechanics remains a limiting factor. STGODE, enhanced with our graph and its Neural ODE-based temporal continuity, emerges as a powerful short-term baseline. Our EISTGNN, however, achieves superior consistency and long-horizon stability through its tightly coupled design: the Spatio-Contact SIR model and the spatiotemporal graph learner mutually constrain each other end-to-end, ensuring representations align with both data and epidemiological principles. Ablation studies (Sections 4.6 and 4.7) quantify the contributions of each component.

4.6. Effects of mechanistic integration

To investigate the impact of the epidemiological component on interpretable epidemic forecasting, we designed two variant models: EISTGNN-MC (EISTGNN-Mechanistically-Constrained) and EISTGNN-NM (EISTGNN-Non-Mechanistic). These variants allow to evaluate the contribution of the epidemiological mechanism constraint integrated into the EISTGNN model by introducing the following differences:

(1) **EISTGNN-MC**: The EISTGNN-MC model retains the epidemiological module, which integrates domain knowledge of epidemic dynamics to constrain and guide the graph learning and forecasting process. This variant leverages theory-derived relationships, to enhance spatiotemporal predictions, by modeling both interregional interactions and temporal patterns, influenced by epidemiological parameters.

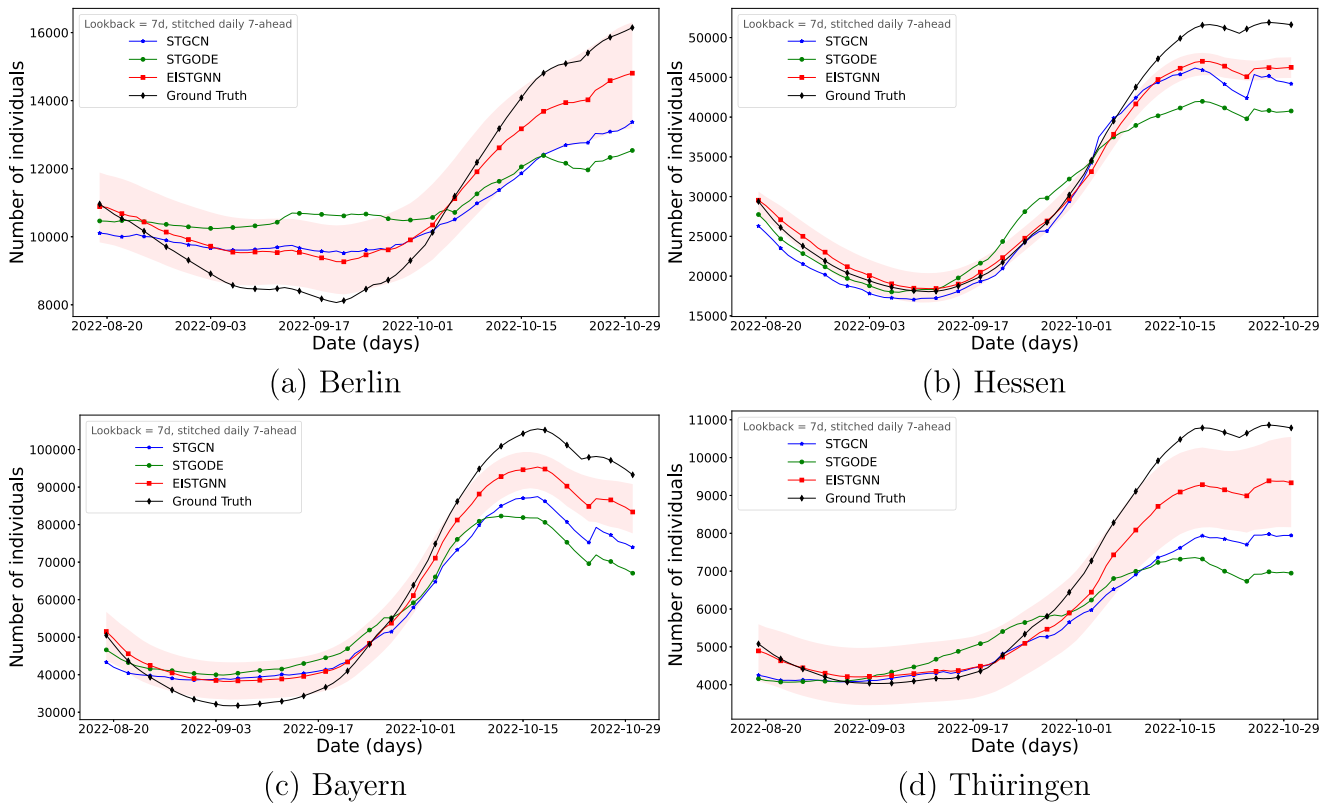

Fig. 7. Visualization of 7-day-ahead forecasts for active cases in representative provinces of Germany. For each calendar date τ, the value $\hat{y}_{\tau|\tau-7}$ represents the prediction made 7 days prior. The continuous curve is formed by connecting these daily predictions across the test period. The ground truth is shown in black, comparisons from other models in color, and the shaded band represents the 95% interval over an ensemble of 5 independent training runs, reflecting prediction uncertainty due to model stochasticity.

Table 5
Effects of mechanistic integration on the China dataset.

	The China dataset									
	L = 7					L = 14				
Model	MAE	RMSE	RAE	PCC	CCC	MAE	RMSE	RAE	PCC	CCC
EISTGNN-NM	44.3 ± 13.2	100.9 ± 39.3	0.3 ± 0.1	97.2 ± 3.0%	96.7 ± 3.1%	71.47 ± 13.72	165.6 ± 36.1	0.4 ± 0.1	93.3 ± 3.4%	92.6 ± 3.4%
EISTGNN-MC	39.6 ± 5.3	83.5 ± 12.1	0.2 ± 0.0	98.3 ± 0.5%	98.1 ± 0.5%	58.5 ± 10.3	147.7 ± 27.2	0.3 ± 0.1	95.4 ± 1.3%	93.8 ± 3.6%
	L = 21					L = 28				
Model	MAE	RMSE	RAE	PCC	CCC	MAE	RMSE	RAE	PCC	CCC
EISTGNN-NM	98.6 ± 21.9	228.6 ± 49.6	0.6 ± 0.1	86.0 ± 6.0%	85.0 ± 5.4%	127.6 ± 43.2	322.4 ± 146.8	0.7 ± 0.3	75.8 ± 14.3%	72.3 ± 15.9%
EISTGNN-MC	91.0 ± 16.6	213.1 ± 31.8	0.5 ± 0.1	88.3 ± 4.7%	84.70 ± 7.7%	111.7 ± 44.1	270.9 ± 100.9	0.6 ± 0.3	75.0 ± 23.8%	73.5 ± 23.0%

(2) **EISTGNN-NM**: The EISTGNN-NM model removes the epidemiological module, excluding the epidemiological constraints and focusing solely on data-driven learning. By eliminating the epidemiological component, this variant relies entirely on spatiotemporal feature extraction through graph learning and the temporal modules, without leveraging domain-specific epidemiological knowledge.

This experimental setup facilitates the isolation and quantification of the impact of mechanistic integration in the EISTGNN model, offering valuable insights into the role of domain knowledge in enhancing epidemic forecasting accuracy.

The experimental results on the China and Germany datasets are shown in Tables 5 and 6, respectively. On the China dataset, EISTGNN-MC consistently outperforms EISTGNN-NM across all forecasting horizons. By incorporating mechanistic constraints, EISTGNN-MC leverages domain-specific epidemiological knowledge to capture complex spatiotemporal dependence more effectively. For instance, at L = 7 and L = 14, EISTGNN-MC achieves notable improvements in MAE and RMSE, while also showing higher PCC and CCC values, demonstrating stronger alignment with the actual epidemic trends. These results emphasize the advantage of integrating mechanistic constraints in achieving more accurate and reliable epidemic forecasts.

Similarly, on the German dataset, EISTGNN-MC demonstrates outstanding performance in both short-term and long-term prediction tasks. For instance, at L = 7, L = 14, and L = 21, EISTGNN-MC achieves substantial reductions in MAE and RMSE compared to EISTGNN-NM, highlighting its robustness in handling extended prediction horizons. In contrast, while EISTGNN-NM performs relatively better at L = 7 compared to other prediction windows, its reliance on purely data-driven learning limits its ability to effectively capture long-term dependence. This limitation results in a pronounced decline in performance as the

Table 6
Effects of mechanistic integration on the Germany dataset.

	The Germany dataset									
	L = 7					L = 14				
Model	MAE	RMSE	RAE	PCC	CCC	MAE	RMSE	RAE	PCC	CCC
EISTGNN-NM	7447.2 ± 0.00	13069.1 ± 0.00	0.4 ± 0.0	90.85 ± 0.0%	82.2 ± 0.0%	8951.3 ± 0.00	159237 ± 0.00	0.5 ± 0.0	84.55 ± 0.0%	70.8 ± 0.0%
EISTGNN-MC	4393.8 ± 416.8	7511.6 ± 184.2	0.2 ± 0.1	97.66 ± 1.3%	94.4 ± 2.5%	7400.7 ± 949.6	12577.1 ± 361.6	0.4 ± 0.1	92.66 ± 2.0%	80.8 ± 11.1%
	L = 21					L = 28				
Model	MAE	RMSE	RAE	PCC	CCC	MAE	RMSE	RAE	PCC	CCC
EISTGNN-NM	8699.4 ± 664.9	14509.7 ± 363.5	0.5 ± 0.1	87.7 ± 4.0%	79.5 ± 6.5%	8914.0 ± 987.4	15017.5 ± 765.7	0.6 ± 0.1	85.6 ± 4.8%	74.3 ± 11.8%
EISTGNN-MC	7487.5 ± 425.9	12220.5 ± 931.9	0.4 ± 0.1	89.0 ± 3.4%	86.0 ± 4.1%	8423.9 ± 422.2	13880.9 ± 1406.2	0.5 ± 0.1	86.2 ± 3.7%	77.0 ± 15.7%

Table 7
Effects of spatiotemporal contact graph on the China dataset.

	The China dataset									
	L = 7					L = 14				
Model	MAE	RMSE	RAE	PCC	CCC	MAE	RMSE	RAE	PCC	CCC
EISTGNN-Static	44.4 ± 11.3	100.0 ± 32.2	0.3 ± 0.1	97.2 ± 2.4%	96.9 ± 2.4%	64.8 ± 8.9	154.2 ± 26.1	0.4 ± 0.1	94.4 ± 1.8%	93.8 ± 1.9%
EISTGNN-SCG	39.6 ± 5.3	83.5 ± 12.1	0.2 ± 0.0	98.3 ± 0.5%	98.1 ± 0.5%	58.5 ± 10.3	147.7 ± 27.2	0.3 ± 0.1	95.4 ± 1.3%	93.8 ± 3.6%
	L = 21					L = 28				
Model	MAE	RMSE	RAE	PCC	CCC	MAE	RMSE	RAE	PCC	CCC
EISTGNN-Static	116.0 ± 42.7	279.4 ± 122.4	0.7 ± 0.2	81.3 ± 10.8%	78.8 ± 12.3%	116.0 ± 31.6	276.8 ± 80.4	0.7 ± 0.2	79.4 ± 8.8%	78.3 ± 9.8%
EISTGNN-SCG	91.0 ± 16.6	213.1 ± 31.8	0.5 ± 0.1	88.3 ± 4.7%	84.7 ± 7.7%	111.7 ± 44.1	270.9 ± 100.9	0.6 ± 0.3	75.0 ± 23.8%	73.5 ± 23.0%

prediction window expands. Overall, the integration of mechanistic constraints empowers EISTGNN-MC to deliver more stable and accurate forecasts, consistently surpassing EISTGNN-NM across diverse datasets and prediction scenarios.

4.7. Effects of spatiotemporal contact graph

To investigate the impact of the Contact Graph Component on spatiotemporal epidemic forecasting, we again devised two variant models: EISTGNN-SCG (EISTGNN with Contact Graph Component) and EISTGNN-Static (EISTGNN with Static Adjacency Matrix). These variants allow us to evaluate the effectiveness of simulating spatiotemporal contact patterns through contact graph learning compared to using a fixed adjacency matrix, and differ as follows:

(1) **EISTGNN-SCG**: This variant incorporates a spatiotemporal contact graph to simulate interregional interaction patterns as a time-varying adjacency matrix. The dynamic adjacency matrix is learned through the graph learning algorithm and represents the evolving interactions between regions over time. By modeling these dynamic changes, this approach aims to approximate real-world interaction flow, allowing the GCN to process accurate and temporally adaptive spatial relationships.
(2) **EISTGNN-Static**: In this variant, the graph structure is represented by a static binary adjacency matrix, where edges are pre-defined based on geographical proximity (1 for neighboring regions and 0 otherwise). While this approach provides a straightforward and computationally efficient alternative, it assumes fixed spatial relationships and does not account for temporal variations in interregional interaction, potentially limiting its ability to adapt to dynamic epidemic patterns.

This experimental setup facilitates the isolation and quantification of the impact of graph learning in EISTGNN, providing insights into the role of simulated temporally adaptive spatial relationships in enhancing epidemic forecasting accuracy.

A comparison of our experimental results for both models operating on the China and Germany datasets, shown in Tables 7 and 8, highlights the impact of graph learning mechanisms on spatiotemporal epidemic forecasting. On both datasets, the EISTGNN-SCG model, which utilizes a graph learning module designed to learn interregional interaction patterns as a dynamic adjacency matrix, consistently outperforms the EISTGNN-Static model. For example, for the China dataset, at L = 14, EISTGNN-SCG achieves an MAE of 58.5 compared to 64.8 for EISTGNN-Static, along with notable improvements in RMSE, PCC, and CCC, demonstrating its ability to effectively model dynamic spatial relationships. Similarly, for the Germany dataset, at L = 28, EISTGNN-SCG achieves a CCC of 77%, significantly higher than the 71.1% achieved by EISTGNN-Static, further validating the benefits of learning and simulating interregional interaction patterns to construct adaptive adjacency matrices.

In contrast, the EISTGNN-Static model, which uses a static binary adjacency matrix based on geographical proximity, performs reasonably well but lacks the flexibility to adapt to temporal variations in spatial relationships. This limitation becomes more apparent in longer forecasting horizons, where the ability to model dynamic and evolving interactions between regions is critical. By incorporating a graph learning mechanism to simulate interregional interaction patterns as a dynamic adjacency matrix, EISTGNN-SCG demonstrates greater adaptability and accuracy. Overall, the use of graph learning enables EISTGNN-SCG to deliver more robust and reliable predictions, highlighting the importance of adaptive graph structures in spatiotemporal epidemic forecasting.

4.8. Interpretability analysis via identified epidemiological parameters

The Spatio-Contact SIR model, as an essential part of our framework, significantly enhances the interpretability of the results generated by our framework. By analyzing the learned epidemic parameters (β, γ, and c) generated by the Spatiotemporal module, we demonstrate how the model captures meaningful interaction patterns between regions, reflecting real-world interaction flow and its spatiotemporal dynamics. We base our analyses on the 7-day ahead prediction results in the best-performing model, further demonstrating how the module enhances the framework's interpretability through its ability to model spatiotemporal interactions.

4.8.1. Contact rates

The contact rates in China exhibit distinct temporal and spatial variations influenced by holidays and interregional interactions. As shown in Fig. 8, intraregional interactions (diagonal elements) remain dominant across all dates, reflecting the prevalence of localized travel.

Table 8
Effects of spatiotemporal contact graph on the Germany dataset.

	The Germany dataset									
	L = 7					L = 14				
Model	MAE	RMSE	RAE	PCC	CCC	MAE	RMSE	RAE	PCC	CCC
EISTGNN-Static	5709.5 ± 2841.3	9294.1 ± 4755.8	0.3 ± 0.2	95.5 ± 3.8%	88.9 ± 11.8%	7480.5 ± 2640.8	12565.5 ± 4267.2	0.4 ± 0.2	91.6 ± 3.5%	80.0 ± 14.5%
EISTGNN-SCG	4393.8 ± 416.8	7511.6 ± 184.2	0.2 ± 0.1	97.7 ± 1.3%	94.4 ± 2.5%	7400.7 ± 949.6	12577.1 ± 361.6	0.4 ± 0.1	92.7 ± 1.9%	80.8 ± 11.1%
	L = 21					L = 28				
Model	MAE	RMSE	RAE	PCC	CCC	MAE	RMSE	RAE	PCC	CCC
EISTGNN-Static	8729.3 ± 2594.2	14413.1 ± 4162.4	0.5 ± 0.2	85.4 ± 4.8%	81.1 ± 7.0%	9230.7 ± 3171.6	15435.2 ± 5000.7	0.5 ± 0.2	85.6 ± 3.4%	71.1 ± 22.1%
EISTGNN-SCG	7487.5 ± 425.9	12220.5 ± 931.9	0.4 ± 0.1	89.0 ± 3.4%	86.0 ± 4.1%	8423.9 ± 422.2	13880.9 ± 1406.2	0.5 ± 0.1	86.2 ± 3.7%	77.0 ± 15.7%

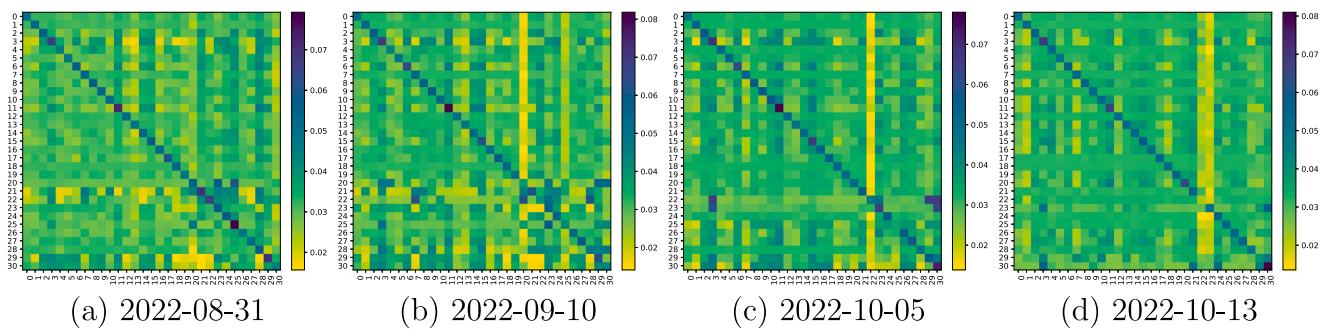

Fig. 8. Temporal Evolution of Learned Contact Rates Across Provinces in China.

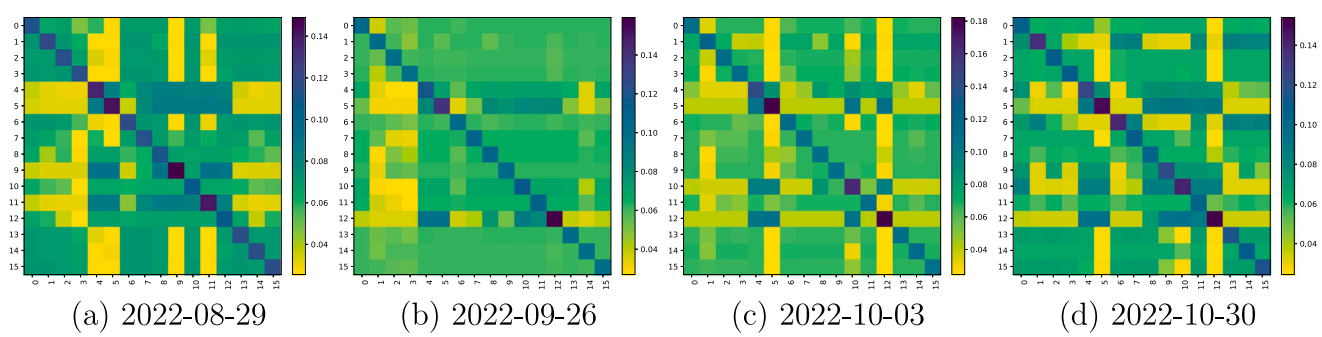

Fig. 9. Temporal Evolution of Learned Contact Rates Across Federal States in Germany.

Off-diagonal elements, however, reveal notable interprovincial interactions during specific periods. On August 31, 2022, marking the end of summer vacations, elevated contact rates are observed between densely populated areas such as Beijingshi (index 0), Tianjinshi (1), and Hebeisheng (2), corresponding to student and family travel. Coastal regions like Zhejiangsheng (10) and Fujian (12) also demonstrate significant interactions, driven by their status as popular tourist destinations. During the *Mid-Autumn Festival* on September 10, 2022, regions with strong familial or cultural ties, such as Guangdongsheng (18) and Hainansheng (20), show increased connectivity, highlighting family reunions. Fig. 8(c) illustrates October 5, 2022, during the *National Day Golden Week* when interprovincial travel peaks, particularly among tourist hubs like Shanghaishi (8), Zhejiangsheng (10), and Guangdongsheng (18). By October 13, 2022, as the holiday season concludes, interregional interactions decline slightly, with economic centers like Shanghaishi (8) and Guangdongsheng (18) sustaining higher contact rates due to commuting and economic activity.

For Germany, the learned contact rates reflect the effects of holidays, cultural events, and regular commuting on cross-region interaction patterns. As shown in Fig. 9, intraregional interactions (diagonal elements) dominate consistently, underscoring the importance of localized movements within states. On August 29, 2022, at the end of summer breaks, Berlin (index 1) and Brandenburg (0) show heightened interactions due to commuting and vacation-related travel. Similarly, coastal regions like Schleswig-Holstein (11) exhibit higher contact rates, driven by their role as popular summer destinations. On September 26, 2022, during the *Oktoberfest* in Bayern (3), interregional interactions rise sharply, particularly between Bayern and neighboring states such as Baden-Württemberg (2) and Hessen (5), reflecting festival-driven population flows. Fig. 9(b) highlights this spike, with darker off-diagonal elements indicating substantial interregional connectivity. October 3, 2022, associated with Germany's Unity Day, shows moderate interregional travel, with populous states like Nordrhein-Westfalen (9) and Rheinland-Pfalz (10) contributing notably to increased connectivity. By October 30, 2022, interactions stabilize,

with economically significant states like Bayern (3) and Nordrhein-Westfalen (9) maintaining higher intraregional activity due to their economic roles and population density.

These findings confirm the critical role of interregional interaction components, driven by holidays, cultural events, and economic activities, in shaping the dynamics of contact rates. The Spatio-Contact SIR model effectively captures these variations, demonstrating its ability to align learned parameters with real-world interregional contact patterns. By identifying both localized and interregional trends, the model enhances interpretability and provides valuable insights for epidemic modeling and public health policy planning.

4.8.2. Effective reproduction number

Given that the parameters (β, γ, c) serve as latent bridging variables in our framework and exhibit inherent identifiability coupling, we focus the interpretability and uncertainty analysis on R_t, which is jointly determined by them. This approach provides a more coherent and decision-relevant mechanistic interpretation. To enhance the interpretability of the proposed Spatio-Contact SIR model, we devised a new definition of the effective reproduction number R_t. Unlike the traditional basic reproduction number R_0, which assumes homogeneous mixing and static conditions, our R_t incorporates spatial heterogeneity and temporal dynamics. This is achieved by integrating learned parameters into a time-dependent next-generation matrix (Diekmann et al., 1990), providing a more flexible and context-sensitive measure of instantaneous transmission potential. In classical epidemic models, R_0 is given by:

$$R_0 = \frac{\beta}{\gamma} \tag{24}$$

where β denotes the infection rate and γ is the recovery rate. This ratio represents the initial growth potential of an outbreak in a fully susceptible population. However, such a definition does not account for interregional interactions or time-varying conditions.

In our framework, transmission patterns vary dynamically across both space and time. To approximate a meaningful R_t, we construct a time-dependent next-generation matrix that incorporates these variations. To this end, we first define a diagonal matrix at time t,

$$D(t) = \operatorname{diag}\left(\frac{\beta_1(t)}{\gamma_1(t)}, \frac{\beta_2(t)}{\gamma_2(t)}, \dots, \frac{\beta_Q(t)}{\gamma_Q(t)}\right), \tag{25}$$

where each element represents the transmission-to-recovery ratio for a specific region. We also define a contact matrix

$$C(t) = (c_{ij}(t)), \quad i, j = 1, 2, \dots, Q, \tag{26}$$

where $c_{ij}(t)$ denotes the contact intensity from region j to region i. The next-generation matrix is then computed as

$$M(t) = D(t)C(t) \tag{27}$$

where $M_{ij}(t)$ represents the expected number of secondary infections in region i caused by a single infected individual in region j, assuming a fully susceptible population. Following the next-generation matrix approach (Diekmann et al., 1990), the effective reproduction number R_t is then determined as the spectral radius (i.e., the largest eigenvalue) of $M(t)$:

$$R_t = \lambda_{\max}(M(t)). \tag{28}$$

This represents the dominant transmission potential in a structured population. This formulation encapsulates both spatial heterogeneity and temporal dynamics, providing a robust and adaptive estimate of the instantaneous transmission potential.

Figs. 10 and 11 illustrate the trajectory of R_t for representative regions of China and Germany. In Fig. 10, the trajectory for Shaanxisheng is shown, where shaded blue intervals correspond to key periods of increased interregional interactions and social gatherings, such as the *Back-to-school Season* and *Golden Week*. During these intervals, R_t rises markedly, reflecting elevated transmission risks due to intensified population mixing. Once these events conclude, R_t declines again, signifying a restoration of lower transmission potential.

Similarly, Fig. 11 highlights the transient increase of R_t in the state of Bayern during Oktoberfest. The shaded blue area indicates the festival timeframe, during which R_t surges as crowd density and contact rates increase markedly. After the event, R_t steadily declines, aligning with a reduction in social mixing and return to stable patterns of reduced movement. These upward and downward shifts in R_t, synchronized with the presence or absence of significant events, demonstrate the model's ability to dynamically capture real-world social and movement variations.

Interestingly, the maximal R_t values in Shaanxisheng are higher than those in Bayern during key events, reflecting an augmented transmission potential in China due to concentrated periods of increased interregional interactions (e.g., *Golden Week* and *Back-to-school Season*). While Shaanxisheng has a larger total population, its population density is relatively similar to that of Bayern, suggesting that interaction patterns are the primary driver of these differences, though the potential influence of population density cannot be ruled out. However, the actual epidemic severity in Germany was greater, influenced by its more relaxed public health policies and a higher cumulative infection rate, which increased the effective susceptibility of the population (Huber and Langen, 2020). This discrepancy highlights the distinction between theoretical transmission potential captured by R_t and the realized epidemic impact shaped by regional policies and long-term dynamics. Within event windows (e.g., Golden Week, back-to-school, Oktoberfest), EISTGNN yields small peak-timing/amplitude discrepancies and the empirical 95% uncertainty bands from five independent runs remain well calibrated, indicating accurate and stable responses to abrupt changes.

The analysis of contact rate and effective reproduction number demonstrates that EISTGNN provides accurate predictions. Meanwhile, the Spatio-Contact SIR component enhances interpretability by uncovering the underlying spatiotemporal trends in population interactions. These results underscore the potential of our framework as a robust tool for epidemic forecasting and policy evaluation.

5. Conclusion

Since the outbreak of COVID-19, accurate epidemic forecasting has emerged as a pivotal area of research. In this study, we introduce a novel framework that integrates mechanistic modeling with deep learning to model the spatiotemporal spread dynamics of infectious diseases. Our hybrid framework is the Epidemiology-informed Spatiotemporal Graph Neural Network (EISTGNN), which incorporates the Spatio-Contact SIR model (SCSIR) to account for both intra-regional homogeneity and inter-regional heterogeneity. The SCSIR model serves as an embedded module, utilizing the contact rates to capture interregional interaction patterns and effectively reflect the complexities of disease transmission. By embedding the SCSIR model into EISTGNN, we seamlessly merge domain knowledge, mechanistic constraints and data-driven techniques. This integration enhances the interpretability of deep learning and ensures that its training is guided by epidemiological knowledge. Furthermore, we incorporate temporal decomposition and spatiotemporal graph learning modules to accurately model interregional interactions and spatiotemporal dependence, enabling the framework to discern complex transmission dynamics and interactions.

We validated the effectiveness of our proposed method using real-world data from China and Germany over the course of a year. This period encompasses a variety of temporal and spatial dynamics, including different seasons and major holidays and feasts in both countries. Comparative experiments against several baseline methods demonstrate that our framework consistently outperforms existing approaches in epidemic forecasting. Additionally, ablation studies highlight the significance of each model component, underscoring the framework's

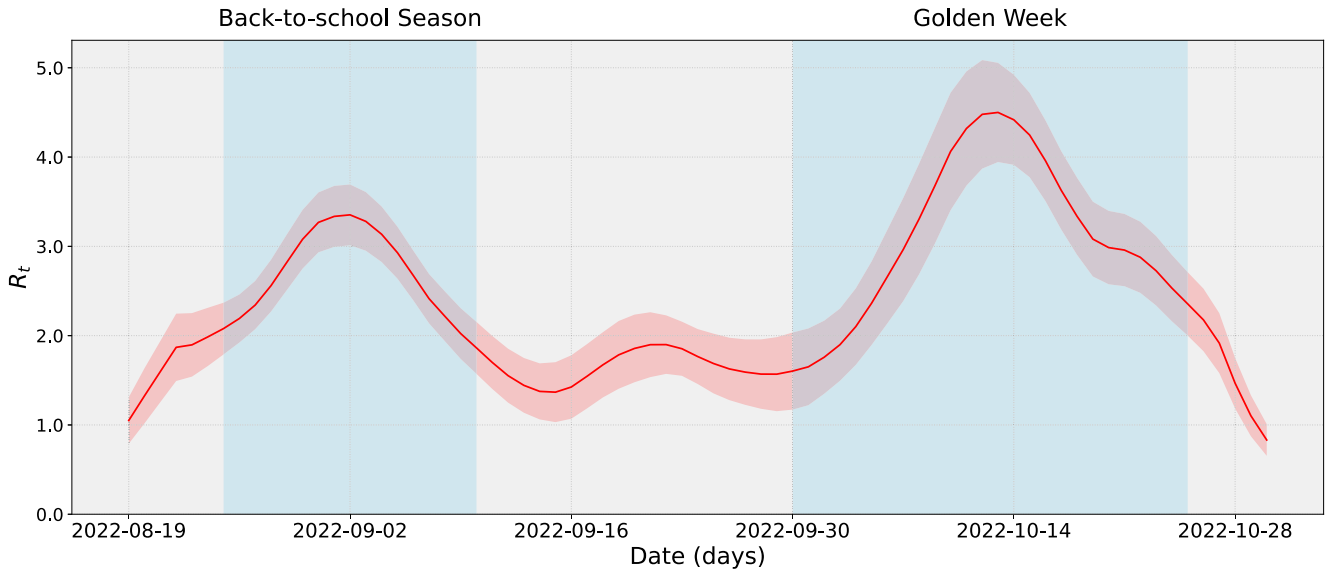

Fig. 10. Visualization of R_t curves in Shaanxisheng of China. The blue shaded areas denote event windows (holidays/mass gatherings). The red band indicates the 95% interval of R_t values derived from an ensemble of 5 independently trained models.

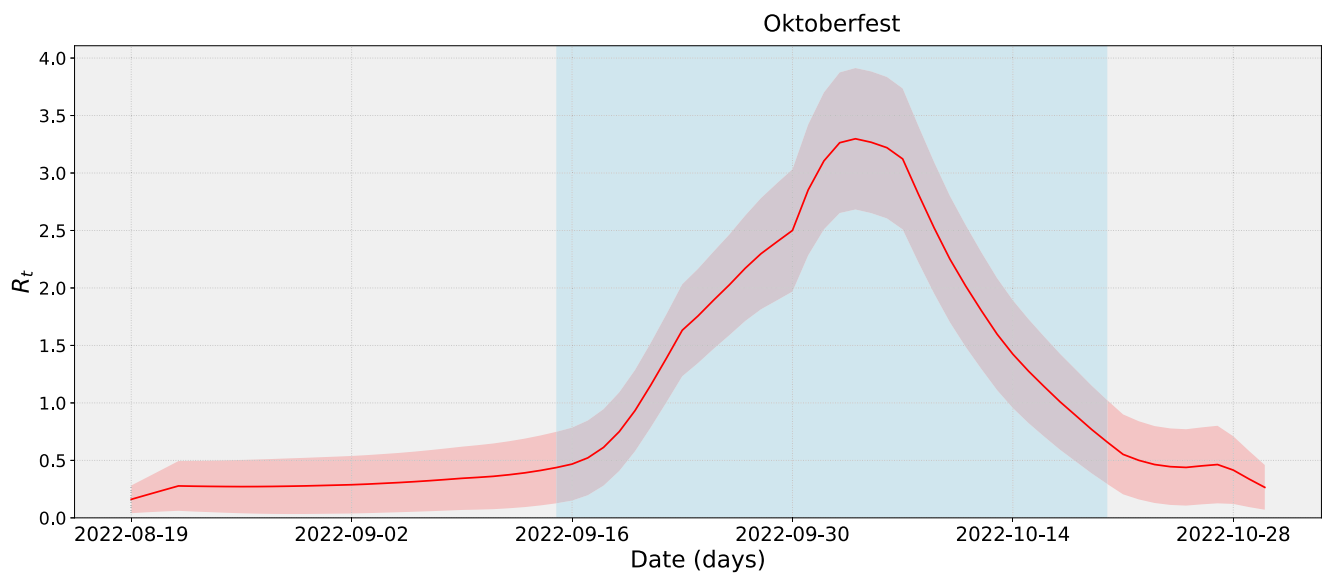

Fig. 11. Visualization of R_t curves in Bayern of Germany. The blue shaded area denote event windows (holidays/mass gatherings). The red band indicates the 95% interval of R_t values derived from an ensemble of 5 independently trained models.

robustness and precision in real-world applications. These results affirm our method's capability to adeptly capture and leverage spatiotemporal patterns for reliable predictions.

Beyond forecasting accuracy, we delve into the analysis of learned epidemiological parameters, such as the contact rate matrix and the effective reproduction number, to better understand disease transmission dynamics in both countries. Our findings indicate that these parameters closely mirror real-world observations, effectively capturing the fundamental patterns of epidemic spread. Overall, our framework not only excels in predictive performance but also provides valuable, interpretable epidemiological insights. This dual advantage demonstrates its potential as a powerful tool for understanding and managing infectious diseases. Future research may extend this integration of mathematical models and neural networks to other domains, further harnessing the synergy between domain-specific knowledge and deep learning.

Furthermore, the modular design of EISTGNN ensures favorable scalability along both spatial and temporal dimensions, outlining a clear path for future large-scale applications. Spatially, the computational architecture is well understood: the mechanistic SCSIR simulation scales quadratically with the number of regions ($O(Q^2)$), while all neural components scale linearly. This scaling behavior is empirically validated, with computational cost increasing predictably and manageably with the number of regions, as demonstrated by the comparative analysis between the Germany ($Q = 16$) and China ($Q = 31$) datasets. Temporally, the use of parallelizable Temporal Convolutional Networks guarantees efficient linear-time processing of long sequences. This capability is substantiated by the model's robust performance in accurate multi-week (up to 28-day) forecasts, confirming its proficiency in handling long-horizon dependencies. Together, these analyses confirm that the EISTGNN framework is computationally feasible for

problems involving hundreds of regions and month-scale horizons. Scaling to even larger networks – a key direction for future work – will build upon this foundation through techniques such as graph sparsification, meta-population modeling, and distributed training.

6. Limitation and future work

While our proposed framework demonstrates strong performance during the evaluation period from October 30, 2021, to October 30, 2022. One of the limitations of our proposed framework is its inability to fully account for the specific epidemic prevention policies implemented within different provinces in China and states in Germany. For instance, during periods when certain regions imposed targeted lockdowns or localized vaccination campaigns, our model may not accurately capture the resulting changes in transmission dynamics. Although our framework outperforms baseline methods under relatively stable conditions, its predictive accuracy may decrease in the presence of such policy-driven variations. Enhancing the model to incorporate regional policy measures represents an important direction for future research.

Regarding uncertainty, we center interpretability and uncertainty reporting on the effective reproduction number R_t, rather than on individual epidemiological parameters, because (β, γ, c) act as latent bridging variables and exhibit identifiability coupling. The reported 95% intervals/bands reflect variability across independent training runs and therefore primarily quantify training stochasticity, not observational noise. Extending the framework to incorporate data-level uncertainty (e.g., observation noise and reporting bias) is a promising avenue for future work.

CRediT authorship contribution statement

Shuai Han: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lukas Stelz:** Writing – review & editing, Funding acquisition, Conceptualization. **Thomas R. Sokolowski:** Writing – review & editing, Formal analysis, Conceptualization. **Kai Zhou:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Horst Stöcker:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

Ali, S.T., Wang, L., Lau, E.H.Y., Xu, X.K., Du, Z., Wu, Y., Leung, G.M., Cowling, B.J., 2020. Serial interval of SARS-CoV-2 was shortened over time by nonpharmaceutical interventions. Science 369 (6507), 1106–1109.

Anderson, R.M., May, R.M., 1992. Infectious Diseases of Humans: Dynamics and Control. Oxford University Press.

Bai, S., Kolter, J., Koltun, V., 2018. An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. arXiv preprint arXiv:1803.01271.

Baker, S.R., Bloom, N., Davis, S.J., Terry, S.J., Yoon, J.H., 2020. The economic impact of COVID-19. J. Econ. Perspect. 34 (4), 3–23.

Blanker, L.D., 2024. DXY-COVID-19-data: Releases. https://github.com/BlankerL/DXY-COVID-19-Data/releases. (Accessed on 09 Septenber 2024).

Box, G.E., Jenkins, G.M., 1976a. Time Series Analysis: Forecasting and Control. Prentice-Hall.

Box, G.E., Jenkins, G.M., 1976b. Time Series Analysis: Forecasting and Control. Holden-day.

Cao, Q., Jiang, R., Yang, C., Fan, Z., Song, X., Shibasaki, R., 2023. Metapopulation graph neural networks: Deep metapopulation epidemic modeling with human mobility. arXiv preprint arXiv:2306.14857.

Cho, K., Van Merrienboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., Bengio, Y., 2014. Learning phrase representations using RNN encoder-decoder for statistical machine translation. In: Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing. EMNLP, Association for Computational Linguistics, pp. 1724–1734.

Davis, R., Kim, S., 2023. Network-based approaches to understanding mobility patterns in disease spread. Netw. Heal. 5 (3), 150–165.

Deng, S., Wang, S., Rangwala, H., Wang, L., Ning, Y., 2019. Graph message passing with cross-location attentions for long-term ILI prediction. arXiv:1912.10202.

Diekmann, O., Heesterbeek, J.A.P., Metz, J.A.J., 1990. On the definition and the computation of the basic reproduction ratio R0 in models for infectious diseases in heterogeneous populations. J. Math. Biol. 28 (4), 365–382.

Fang, Z., Long, Q., Song, G., Xie, K., 2021. Spatial-temporal graph ODE networks for traffic flow forecasting. In: Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining. ACM, pp. 364–373.

Fisman, D.N., Sabin, M.L., Brauer, F., Fiedler, I., Kamel, A.A., Sale, H.J., 2020. Subclinical SARS-CoV-2 infections and transmission in households: A longitudinal analysis. Lancet Infect. Dis. 20 (8), e246–e251.

Garcia, L., Martinez, A., 2021. Incorporating spatial heterogeneity into epidemiological models. Spat. Heal. 8 (2), 89–101.

Goodfellow, I., Bengio, Y., Courville, A., 2016. Deep Learning. MIT Press.

Han, S., Stelz, L., Stoecker, H., Wang, L., Zhou, K., 2024. Approaching epidemiological dynamics of COVID-19 with physics-informed neural networks. J. Franklin Inst. 361 (6), 106671.

Heesterbeek, H., Anderson, R.M., Andreasen, V., et al., 2015. Modeling infectious disease dynamics in the complex landscape of global health. Science 347 (6227), aaa4339.

Hochreiter, S., Schmidhuber, J., 1997. Long short-term memory. Neural Comput. 9 (8), 1735–1780.

Hsiang, S., Allen, D., Annan-Phan, S., Bell, B., Bolliger, I., Chong, T., et al., 2020. The effect of large-scale anti-contagion policies on the COVID-19 pandemic. Nature 584 (7820), 262–267.

Huang, J., Xu, Z., Lin, L., Liu, C., Wang, L., 2020. Adaptively aggregated graph neural networks for weakly-supervised disease progression modeling. IEEE Trans. Med. Imaging 39 (8), 2624–2635.

Huber, M., Langen, H., 2020. The impact of response measures on COVID-19-related hospitalization and death rates in Germany and Switzerland. Heal. Policy 125 (8), 946–955.

Johnson, E., Lee, M., 2019. Limitations of the SIR model in heterogeneous populations. Int. J. Public Heal. 22 (3), 112–123.

Karniadakis, G.E., Kevrekidis, I.G., Lu, L., et al., 2021. Physics-informed machine learning. Nat. Rev. Phys. 3 (6), 422–440.

Kermack, W., McKendrick, A., 1927. A contribution to the mathematical theory of epidemics. Proc. R. Soc. Lond. Ser. A. Mathematical Phys. Sci. 115 (772), 700–721.

Kraemer, M.U., Yang, C.-H., Gutierrez, B., Wu, C.-H., Klein, B., Pigott, D.M., Group, O.C.-.D.W., du Plessis, L., Faria, N.R., Li, R., et al., 2020. The effect of human mobility and control measures on the COVID-19 epidemic in China. Science 368 (6490), 493–497.

LeCun, Y., Bengio, Y., Hinton, G., 2015. Deep learning. Nature 521 (7553), 436–444.

Lee, A., Park, D., 2023. Machine learning techniques for integrating spatial data in epidemiological models. Comput. Epidemiol 12 (4), 300–315.

Li, Y., Yu, R., Shahabi, C., Liu, Y., 2018. Diffusion convolutional recurrent neural network: Data-driven traffic forecasting. arXiv preprint arXiv:1707.01926.

Li, X., et al., 2022. DSTAGNN: Dynamic spatial-temporal aware graph neural networks for traffic forecasting. IEEE Trans. Intell. Transp. Syst..

Lim, B., Ziegler, S.O., Hoyer, P., Aghabozorgi, S., Loeff, N., Bartek, J., Ghaffari, M., 2020. Temporal fusion transformers for interpretable multi-horizon time series forecasting. Int. J. Forecast. 36 (4), 1808–1826.

Lipton, Z.C., 2018. The mythos of model interpretability. ACM Queue 16 (3), 31–57.

Mao, J., Han, Y., Tanaka, G., Wang, B., 2024. Backbone-based dynamic spatio-temporal graph neural network for epidemic forecasting. Knowl.-Based Syst. 296, 111952.

Miller, S., Chen, W., 2022. Metapopulation models for regional disease interactions. J. Dis. Dyn. 10 (1), 45–60.

Nowak, M.A., May, R.M., 1999. Vaccination and herd immunity for SIR epidemics. Math. Biosci. 149 (1–2), 1–21.

Peeri, N.C., Shrestha, N., Rahman, M.S., Zaki, R., Tan, Z., Bibi, S., Baghbanzadeh, M., Aghamohammadi, N., Zhang, W., Haque, U., 2020. The SARS, MERS and novel coronavirus (COVID-19) epidemics, the newest and biggest global health threats: what lessons have we learned? Int. J. Epidemiol. 49 (3), 717–726.

Qiu, M., Tan, Z., Bao, B.-K., 2024. MSGNN: Multi-scale spatio-temporal graph neural network for epidemic forecasting. Data Min. Knowl. Discov. 38 (4), 2348–2376. http://dx.doi.org/10.1007/s10618-024-01035-w.

Rackauckas, C., Ma, Y., et al., 2020. Universal differential equations for scientific machine learning. arXiv preprint arXiv:2001.04385.

Raissi, M., Perdikaris, P., Karniadakis, G.E., 2019. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. J. Comput. Phys. 378, 686–707.

Rivieccio, B.A., Micheletti, A., Maffeo, M., Zignani, M., Comunian, A., Nicolussi, F., Salini, S., Manzi, G., Auxilia, F., Giudici, M., et al., 2021. CoVid-19, learning from the past: A wavelet and cross-correlation analysis of the epidemic dynamics looking to emergency calls and Twitter trends in Italian lombardy region. PLoS One 16 (2), e0247854.

Robert Koch-Institut, 2024. SARS-CoV-2 infektionen in deutschland. http://dx.doi.org/10.5281/zenodo.4681153, Zenodo record for the RKI dataset.

Rudin, C., 2019. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. Nat. Mach. Intell. 1 (5), 206–215.

Rumelhart, D.E., Hinton, G.E., Williams, R.J., 1986. Learning representations by back-propagating errors. Nature 323 (6088), 533–536.

Smith, J., Doe, J., 2020. Modeling infectious disease spread with homogeneous populations. J. Epidemiological Model. 15 (4), 234–245.

Smola, A.J., Schölkopf, B., 1998. A tutorial on support vector regression. Stat. Comput. 10 (3), 199–222.

Sun, C., Li, C., Lin, X., Zheng, T., Meng, F., Rui, X., Wang, Z., 2023. Attention-based graph neural networks: a survey. Artif. Intell. Rev. 56, 2263–2310.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., Polosukhin, I., 2017. Attention is all you need. Adv. Neural Inf. Process. Syst. 30.

Wallinga, J., Teunis, P., 2006. Estimating the serial interval of SARS-CoV-2 infections. Am. J. Epidemiol. 172 (9), 993–998.

Wang, H., Chen, J., Pan, T., Dong, Z., Zhang, L., Jiang, R., Song, X., 2024. STGformer: Efficient spatiotemporal graph transformer for traffic forecasting. arXiv:2410.00385.

Wang, L., Han, S., Stoecker, H., Zhou, K., Jiang, Y., 2023. Applications of deep learning in forecasting COVID-19 pandemic and county-level risk warning. In: Mathematical Modelling, Simulations, and AI for Emergent Pandemic Diseases. Elsevier, pp. 119–132.

Wu, J.T., Leung, K., Leung, G.M., 2020a. Dynamic spatiotemporal analysis of population flow data to predict the spread of COVID-19. Lancet 395 (10231), 689–697.

Wu, Z., Pan, S., Chen, F., Long, G., Zhang, C., Yu, P.S., 2020b. A comprehensive survey on graph neural networks. IEEE Trans. Neural Netw. Learn. Syst. 32 (1), 4–24.

Wu, Z., Pan, S., Long, G., Jiang, J., Chang, X., Zhang, C., 2020c. Connecting the dots: Multivariate time series forecasting with graph neural networks. In: Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining. pp. 753–763.

Wu, Z., Pan, S., Long, G., Jiang, J., Zhang, C., 2019. Graph WaveNet for deep spatial-temporal graph modeling. arXiv:1906.00121.

Yang, G., Rahman, M., Deng, X., Lin, S., 2021a. Machine learning for COVID-19 forecasting: A systematic review. IEEE Access 9, 18738–18754.

Yang, M., Zhou, M., Kalander, M., Huang, Z., King, I., 2021b. Discrete-time temporal network embedding via implicit hierarchical learning in hyperbolic space. In: Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining. ACM, pp. 1975–1985.

Yang, A., et al., 2021c. EpiGNN: A graph neural network for epidemic forecasting with static transmission risk. IEEE Trans. Biomed. Eng..

Ye, J., Liu, Z., Du, B., Sun, L., Li, W., Fu, Y., Xiong, H., 2022. Learning the evolutionary and multi-scale graph structure for multivariate time series forecasting. In: Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining. ACM, pp. 2296–2306.

Yu, B., Yin, H., Zhu, Z., 2018. Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting. arXiv preprint arXiv:1709.04875.

Zhang, W., Zhou, L., Li, M., Chen, H., 2022. Deep learning approaches for predicting COVID-19 case trends: A comparative study. IEEE Trans. Neural Netw. Learn. Syst. 33 (5), 1800–1812.